

Vector Calculus Lecture Notes

Compiled for Exam Preparation

June 2026

Contents

1	Lecture 01: Concept of Scalar and Vector Point Function	2
1.1	Prerequisites	2
1.1.1	Position Vectors	2
1.1.2	Vector Multiplication	2
1.2	1. Scalar Point Functions	2
1.2.1	Example	2
1.3	2. Vector Point Functions	2
1.3.1	Example	3
1.4	What Comes Next	3
2	Lecture 02: Vector Differentiation	4
2.1	Prerequisites	4
2.2	1. Vector Differential Coefficients	4
2.3	2. Ordinary Vector Differentiation	4
2.4	3. Solved Problems	4
2.4.1	Problem 1: Derivative of a Vector	4
2.4.2	Problem 2: Dot Product Differentiation	5
2.4.3	Problem 3: Exponential Derivative Identity	5
2.4.4	Problem 4: Trigonometric Derivative Identity	5
2.4.5	Problem 5: Cross Product Differentiation	5
2.4.6	Problem 6: Unit Vector Identity	6
2.4.7	Problem 7: Vector Cross Derivative Identity	6
2.4.8	Problem 8: Position Vector and Kinematics	6
2.5	4. Practice Problems	7
2.6	What Comes Next	7
3	Lecture 03: Concept of Gradient and Geometric Meaning	8
3.1	Prerequisites	8
3.2	1. The Del Operator and Gradient	8
3.2.1	The Del Operator (∇)	8
3.2.2	The Gradient ($\text{grad } \phi$)	8
3.2.3	Geometric Meaning	8
3.3	2. Properties of Gradient (With Proofs)	8
3.3.1	Property I: Gradient of Sum/Difference	8
3.3.2	Property II: Gradient of Product	9
3.3.3	Property III: Gradient of Quotient	9
3.3.4	Property IV: Operator Action on Position Vector	9
3.4	3. Solved Problems	9
3.4.1	Problem 1: Basic Gradient Calculation	9
3.4.2	Problem 2: Gradient Evaluation at a Point	10
3.4.3	Problem 3: Gradient of a Scalar Triple Product	10

3.4.4	Problem 4: Gradient of r^n	10
3.4.5	Problem 5: Cross Product of Gradient of $f(r)$ and Position Vector	11
3.5	4. Homework Problems	11
3.6	What Comes Next	11
4	Lecture 04: Concept of Directional Derivative	12
4.1	Prerequisites	12
4.2	1. Directional Derivative	12
4.2.1	Maximum Directional Derivative	12
4.2.2	Unit Normal Vector	12
4.3	2. Solved Problems	12
4.3.1	Problem 1: Basic Directional Derivative	12
4.3.2	Problem 2: Direction Defined by Points	13
4.3.3	Problem 3: Finding Unknown Constants	13
4.3.4	Problem 4: Angle Between Surfaces	14
4.3.5	Problem 5: Combined Concepts	14
4.4	3. Homework Problems	15
4.5	What Comes Next	15
5	Lecture 05: Concept of Curl and Divergence	16
5.1	Prerequisites	16
5.2	1. Divergence of a Vector	16
5.2.1	Solenoidal Vectors	16
5.3	2. Curl of a Vector	16
5.3.1	Irrotational Vectors	16
5.4	3. Important Vector Identities	16
5.4.1	Identity 1: Divergence of Curl is Zero	16
5.4.2	Identity 2: Curl of Curl	17
5.5	4. Solved Problems	17
5.5.1	Problem 1: Solenoidal Condition	17
5.5.2	Problem 2: Divergence at a Point	17
5.5.3	Problem 3: Properties of Position Vector	17
5.5.4	Problem 4: Finding Divergence and Curl	17
5.5.5	Problem 5: Curl of Exponential Field	18
5.5.6	Problem 6: Evaluating Irrotational Flow	18
5.5.7	Problem 7: Curl of a Cross Product Identity	18
5.5.8	Problem 8: Operations on $r^n \vec{r}$	18
5.6	What Comes Next	19
6	Lecture 06: Concept of Vector Integration and Line Integration	20
6.1	Prerequisites	20
6.2	1. Integration of Vector Functions	20
6.2.1	Indefinite Integral	20
6.2.2	Definite Integral	20
6.3	2. Solved Problems (Vector Integration)	20
6.3.1	Problem 1: Basic Definite Integral	20
6.3.2	Problem 2: Finding Velocity and Displacement	21
6.3.3	Problem 3: Integration Involving Dot Product	21
6.4	3. Line Integration	21
6.4.1	How to Evaluate Line Integrals	21
6.5	4. Solved Problems (Line Integration)	22
6.5.1	Problem 4: Line Integration on a Parametric Curve	22
6.5.2	Problem 5: Line Integration on a Parabola	22
6.5.3	Problem 6: Line Integration on an Arbitrary Path	22

6.5.4	Problem 7: Circulation Around a Rectangle (1)	22
6.5.5	Problem 8: Circulation Around a Rectangle (2)	23
6.5.6	Problem 9: Line Integration on a Helix	23
6.6	Key Takeaways	23
6.7	What Comes Next	23
7	Lecture 07: Concept of Surface and Volume Integral (Part 1)	24
7.1	Prerequisites	24
7.2	1. Surface Integration	24
7.2.1	Finding the Unit Normal Vector	24
7.2.2	Projecting the Surface	24
7.3	2. Volume Integration	24
7.3.1	Setting Limits	25
7.4	3. Solved Problems	25
7.4.1	Problem 1: Surface Integral Over a Plane	25
7.5	Key Takeaways	26
7.6	What Comes Next	26
8	Lecture 08: Concept of Surface and Volume Integral (Part 2)	27
8.1	Prerequisites	27
8.2	1. Solved Problems	27
8.2.1	Problem 1 (Revisited): Projection on the yz -plane	27
8.2.2	Problem 2: Surface Integral Over a Sphere	28
8.3	Key Takeaways	29
8.4	What Comes Next	29
9	Lecture 09: Concept of Surface and Volume Integral (Part 3)	30
9.1	Prerequisites	30
9.2	Solved Problems	30
9.2.1	Problem 1: Surface Integral Over a Closed Cube	30
9.2.2	Problem 2: Volume Integral Over a Solid Cylinder	31
9.3	Key Takeaways	32
9.4	What Comes Next	32
10	Lecture 10: Concept of Stokes' Theorem (Part 1)	33
10.1	Prerequisites	33
10.2	Stokes' Theorem	33
10.2.1	Statement	33
10.2.2	Orientation (The Right-Hand Screw Rule)	33
10.2.3	Why is this Theorem Useful?	33
10.3	Solved Problems	33
10.3.1	Problem 1: Verifying Stokes' Theorem on a Rectangle	33
10.4	What Comes Next	35
11	Lecture 11: Concept of Stokes' Theorem (Part 2)	36
11.1	Prerequisites	36
11.2	Solved Problems	36
11.2.1	Problem 1: Verifying Stokes' Theorem on a Unit Circle	36
11.2.2	Problem 2: Verifying Stokes' Theorem on a Hemisphere	37
11.2.3	Practice Application	38
11.3	What Comes Next	38
12	Lecture 12: Concept of Gauss Divergence Theorem (Part 1)	39
12.1	Prerequisites	39
12.2	Gauss Divergence Theorem	39

12.2.1	Statement	39
12.2.2	Why is this Theorem Useful?	39
12.3	Solved Problems	39
12.3.1	Problem 1: Integration over a Tetrahedron	39
12.3.2	Problem 2: Integration over a Sphere	40
12.4	What Comes Next	41
13	Lecture 13: Concept of Gauss Divergence Theorem (Part 2)	42
13.1	Prerequisites	42
13.2	Solved Problems	42
13.2.1	Problem 1: Verifying Gauss Divergence Theorem on a Cylinder	42
13.2.2	Problem 2: Center of Gravity Trick for Cones	43
13.3	What Comes Next	44
14	Lecture 14: Concept of Green's Theorem	45
14.1	Prerequisites	45
14.2	Green's Theorem in a Plane	45
14.2.1	Statement	45
14.2.2	Why is this Theorem Useful?	45
14.3	Solved Problems	45
14.3.1	Problem 1: Evaluating a Line Integral via Double Integral	45
14.3.2	Problem 2: Verifying Green's Theorem on a Circle	46
14.3.3	Problem 3: Verifying Green's Theorem between Curves	47
14.4	Conclusion	47

1 Lecture 01: Concept of Scalar and Vector Point Function

Series: Vector Calculus **Lecture:** 01 of 14 **Video:** <https://www.youtube.com/watch?v=Y4ZFBx4OuAE>

1.1 Prerequisites

Before diving into vector calculus, let's recall basic vector algebra from high school. A **scalar** is a quantity with only magnitude (e.g., speed, number). A **vector** has both magnitude and direction (e.g., velocity).

1.1.1 Position Vectors

In a 3D Cartesian coordinate system (X, Y, Z) with unit vectors $\hat{i}, \hat{j}, \hat{k}$, we represent a point A in 3D space with a **position vector** starting from the origin O to point A . We denote this vector as $\vec{OA}$.

1.1.2 Vector Multiplication

We have two ways to multiply vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$: 1. **Scalar Product (Dot Product)**: The result is a scalar number.

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

2. **Vector Product (Cross Product)**: The result is a new vector perpendicular to both original vectors.

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

1.2 1. Scalar Point Functions

Imagine a region or domain R in 3D space where we can pick any point (x, y, z) .

A **scalar point function** maps a 3D point from this domain to a single real number (a scalar). The output is a 1D scalar value.

Block Diagram Representation: [Input (3D point)] $\rightarrow$ [Scalar Point Function] $\rightarrow$ [Scalar (Real Number)]

1.2.1 Example

Consider the scalar point function:

$$f(x, y, z) = xy + x^2z$$

Notice there are no unit vectors $(\hat{i}, \hat{j}, \hat{k})$ in the expression. Let us evaluate this function at the specific point $(1, 2, -1)$:

$$f(1, 2, -1) = (1)(2) + (1)^2(-1) = 2 - 1 = 1$$

The output is 1, which is a pure scalar number.

Other examples of scalar point functions: * $f(x, y, z) = xyz^2$ * $f(x, y, z) = xyz + x^2y^2$

1.3 2. Vector Point Functions

A **vector point function** maps a 3D point from the domain to a new vector in 3D space.

Because the output is a vector, the function rule will always contain the unit vectors $\hat{i}, \hat{j}$, and $\hat{k}$.

Block Diagram Representation: [Input (3D point)] $\rightarrow$ [Vector Point Function] $\rightarrow$ [Vector]

1.3.1 Example

Consider the vector point function:

$$\vec{F}(x, y, z) = xy\hat{i} + x^2z\hat{j} + y\hat{k}$$

Let us evaluate this function at the same point $(1, 2, -1)$:

$$\vec{F}(1, 2, -1) = (1)(2)\hat{i} + (1)^2(-1)\hat{j} + (2)\hat{k}$$

Simplifying the expression gives:

$$\vec{F}(1, 2, -1) = 2\hat{i} - \hat{j} + 2\hat{k}$$

The output is a vector, possessing both magnitude and direction.

Other examples of vector point functions: * $\vec{F}(x, y, z) = x^2\hat{i} + yz\hat{j} + z^2\hat{k}$ * $\vec{F}(x, y, z) = x\hat{i} + y^2\hat{j} + xyz\hat{k}$

1.4 What Comes Next

2 Lecture 02: Vector Differentiation

Series: Vector Calculus **Lecture:** 02 of 14 **Video:** <https://www.youtube.com/watch?v=t-xP272h8Xc>

2.1 Prerequisites

2.2 1. Vector Differential Coefficients

When a vector point function $\vec{F}$ depends on multiple variables like x, y , and z , we use partial differentiation. The derivatives we find are called **Vector Differential Coefficients**.

Let us take an example vector field function:

$$\vec{F}(x, y, z) = x^2\hat{i} + y^2z\hat{j} + z\hat{k}$$

We can differentiate this partially with respect to x :

$$\frac{\partial \vec{F}}{\partial x} = 2x\hat{i} + 0\hat{j} + 0\hat{k} = 2x\hat{i}$$

Similarly, with respect to y and z :

$$\begin{aligned}\frac{\partial \vec{F}}{\partial y} &= 2yz\hat{j} \\ \frac{\partial \vec{F}}{\partial z} &= y^2\hat{j} + \hat{k}\end{aligned}$$

Notice that the result of differentiating a vector point function is always another vector point function.

2.3 2. Ordinary Vector Differentiation

If a vector function depends on a single scalar parameter, like time t , we use ordinary differentiation notation: $\frac{d\vec{r}}{dt}$. Example: $\vec{F}(t) = t^2\hat{i} + (t^2 + 3)\hat{j} + t^3\hat{k}$

- **Velocity:** The first derivative gives the velocity vector $\vec{v}$.

$$\vec{v} = \frac{d\vec{F}}{dt} = 2t\hat{i} + 2t\hat{j} + 3t^2\hat{k}$$

- **Acceleration:** The second derivative gives the acceleration vector $\vec{a}$.

$$\vec{a} = \frac{d^2\vec{F}}{dt^2} = 2\hat{i} + 2\hat{j} + 6t\hat{k}$$

2.4 3. Solved Problems

2.4.1 Problem 1: Derivative of a Vector

Question: If $\vec{r} = a \cos t\hat{i} + a \sin t\hat{j} + t\hat{k}$, find $\frac{d\vec{r}}{dt}$, $\frac{d^2\vec{r}}{dt^2}$ and $\left|\frac{d^2\vec{r}}{dt^2}\right|$.

Solution: First, find velocity by differentiating once:

$$\frac{d\vec{r}}{dt} = -a \sin t\hat{i} + a \cos t\hat{j} + \hat{k}$$

Next, find acceleration by differentiating again:

$$\frac{d^2\vec{r}}{dt^2} = -a \cos t\hat{i} - a \sin t\hat{j}$$

Finally, find the magnitude of the acceleration:

$$\left|\frac{d^2\vec{r}}{dt^2}\right| = \sqrt{(-a \cos t)^2 + (-a \sin t)^2 + 0^2} = \sqrt{a^2(\cos^2 t + \sin^2 t)} = a$$

2.4.2 Problem 2: Dot Product Differentiation

Question: If $\vec{u} = t^2\hat{i} - t\hat{j} + (2t+1)\hat{k}$ and $\vec{v} = (2t-3)\hat{i} + \hat{j} - t\hat{k}$, find $\frac{d}{dt}(\vec{u} \cdot \vec{v})$ at $t = 1$.

Solution: First, compute the dot product algebraically:

$$\begin{aligned}\vec{u} \cdot \vec{v} &= (t^2)(2t-3) + (-t)(1) + (2t+1)(-t) \\ \vec{u} \cdot \vec{v} &= 2t^3 - 3t^2 - t - 2t^2 - t = 2t^3 - 5t^2 - 2t\end{aligned}$$

Now, differentiate the result with respect to t :

$$\frac{d}{dt}(\vec{u} \cdot \vec{v}) = \frac{d}{dt}(2t^3 - 5t^2 - 2t) = 6t^2 - 10t - 2$$

Evaluate at $t = 1$:

$$\left. \frac{d}{dt}(\vec{u} \cdot \vec{v}) \right|_{t=1} = 6(1)^2 - 10(1) - 2 = 6 - 10 - 2 = -6$$

2.4.3 Problem 3: Exponential Derivative Identity

Question: If $\vec{r} = \vec{a}e^{nt} + \vec{b}e^{-nt}$, where $\vec{a}$ and $\vec{b}$ are constant vectors, prove that $\frac{d^2\vec{r}}{dt^2} - n^2\vec{r} = 0$.

Solution: Compute the first derivative:

$$\frac{d\vec{r}}{dt} = n\vec{a}e^{nt} - n\vec{b}e^{-nt}$$

Compute the second derivative:

$$\frac{d^2\vec{r}}{dt^2} = n^2\vec{a}e^{nt} + (-n)^2\vec{b}e^{-nt} = n^2(\vec{a}e^{nt} + \vec{b}e^{-nt})$$

Notice the term inside the brackets is $\vec{r}$:

$$\frac{d^2\vec{r}}{dt^2} = n^2\vec{r} \implies \frac{d^2\vec{r}}{dt^2} - n^2\vec{r} = 0 \quad (\text{Proved})$$

2.4.4 Problem 4: Trigonometric Derivative Identity

Question: If $\vec{r} = (\cos nt)\hat{i} + (\sin nt)\hat{j}$, prove that $\frac{d^2\vec{r}}{dt^2} = -n^2\vec{r}$.

Solution: First derivative:

$$\frac{d\vec{r}}{dt} = -n\sin nt\hat{i} + n\cos nt\hat{j}$$

Second derivative:

$$\frac{d^2\vec{r}}{dt^2} = -n^2\cos nt\hat{i} - n^2\sin nt\hat{j} = -n^2(\cos nt\hat{i} + \sin nt\hat{j}) = -n^2\vec{r} \quad (\text{Proved})$$

2.4.5 Problem 5: Cross Product Differentiation

Question: If $\vec{r} = (5t^2)\hat{i} + t\hat{j} - t^3\hat{k}$ and $\vec{s} = (\sin t)\hat{i} - (\cos t)\hat{j}$, find the value of $\frac{d}{dt}(\vec{r} \times \vec{s})$.

Solution: First, compute the cross product determinant:

$$\vec{r} \times \vec{s} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5t^2 & t & -t^3 \\ \sin t & -\cos t & 0 \end{vmatrix}$$

$$\vec{r} \times \vec{s} = (-t^3\cos t)\hat{i} - (t^3\sin t)\hat{j} + (-5t^2\cos t - t\sin t)\hat{k}$$

Take the derivative using product rule on each component:

$$\frac{d}{dt}(\vec{r} \times \vec{s}) = (-3t^2\cos t + t^3\sin t)\hat{i} - (3t^2\sin t + t^3\cos t)\hat{j} - (10t\cos t - 5t^2\sin t + \sin t + t\cos t)\hat{k}$$

Simplify:

$$\frac{d}{dt}(\vec{r} \times \vec{s}) = (t^3\sin t - 3t^2\cos t)\hat{i} - (3t^2\sin t + t^3\cos t)\hat{j} - (11t\cos t - 5t^2\sin t + \sin t)\hat{k}$$

2.4.6 Problem 6: Unit Vector Identity

Question: Show that $\vec{R} \times d\vec{R} = \frac{\vec{r} \times d\vec{r}}{r^2}$, where $\vec{R} = \frac{\vec{r}}{r}$ (unit vector).

Solution: Differentiate $\vec{R}$:

$$d\vec{R} = d\left(\frac{\vec{r}}{r}\right) = d\left(\frac{1}{r}\vec{r}\right) = \frac{1}{r}d\vec{r} + d\left(\frac{1}{r}\right)\vec{r} = \frac{1}{r}d\vec{r} - \frac{1}{r^2}dr\vec{r}$$

Now compute the cross product $\vec{R} \times d\vec{R}$:

$$\vec{R} \times d\vec{R} = \left(\frac{\vec{r}}{r}\right) \times \left(\frac{1}{r}d\vec{r} - \frac{dr}{r^2}\vec{r}\right) = \frac{\vec{r} \times d\vec{r}}{r^2} - \frac{dr}{r^3}(\vec{r} \times \vec{r})$$

Since the cross product of any vector with itself is zero ($\vec{r} \times \vec{r} = 0$), we have:

$$\vec{R} \times d\vec{R} = \frac{\vec{r} \times d\vec{r}}{r^2} \quad (\text{Proved})$$

2.4.7 Problem 7: Vector Cross Derivative Identity

Question: If $\frac{d\vec{a}}{dt} = \vec{r} \times \vec{a}$ and $\frac{d\vec{b}}{dt} = \vec{r} \times \vec{b}$, show that $\frac{d}{dt}(\vec{a} \times \vec{b}) = \vec{r} \times (\vec{a} \times \vec{b})$.

Solution: Apply the derivative product rule for cross products:

$$\frac{d}{dt}(\vec{a} \times \vec{b}) = \frac{d\vec{a}}{dt} \times \vec{b} + \vec{a} \times \frac{d\vec{b}}{dt}$$

Substitute the given relationships:

$$\frac{d}{dt}(\vec{a} \times \vec{b}) = (\vec{r} \times \vec{a}) \times \vec{b} + \vec{a} \times (\vec{r} \times \vec{b})$$

Using the vector triple product identity $\vec{x} \times (\vec{y} \times \vec{z}) = (\vec{x} \cdot \vec{z})\vec{y} - (\vec{x} \cdot \vec{y})\vec{z}$: * First term: $(\vec{r} \times \vec{a}) \times \vec{b} = -\vec{b} \times (\vec{r} \times \vec{a}) = -[(\vec{b} \cdot \vec{a})\vec{r} - (\vec{b} \cdot \vec{r})\vec{a}] = (\vec{r} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{b})\vec{r}$ * Second term: $\vec{a} \times (\vec{r} \times \vec{b}) = (\vec{a} \cdot \vec{b})\vec{r} - (\vec{a} \cdot \vec{r})\vec{b}$

Add the two terms:

$$\frac{d}{dt}(\vec{a} \times \vec{b}) = (\vec{r} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{b})\vec{r} + (\vec{a} \cdot \vec{b})\vec{r} - (\vec{a} \cdot \vec{r})\vec{b} = (\vec{r} \cdot \vec{b})\vec{a} - (\vec{r} \cdot \vec{a})\vec{b}$$

By the vector triple product definition, this expands exactly to:

$$\vec{r} \times (\vec{a} \times \vec{b}) = (\vec{r} \cdot \vec{b})\vec{a} - (\vec{r} \cdot \vec{a})\vec{b}$$

Hence:

$$\frac{d}{dt}(\vec{a} \times \vec{b}) = \vec{r} \times (\vec{a} \times \vec{b}) \quad (\text{Proved})$$

2.4.8 Problem 8: Position Vector and Kinematics

Question: A particle moves so that its position vector is given by $\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$, where ω is a constant. Show that: 1. The velocity $\vec{v}$ of the particle is perpendicular to $\vec{r}$. 2. The acceleration $\vec{a}$ is directed towards the origin and has magnitude proportional to the distance from the origin. 3. $\vec{r} \times \vec{v}$ is a constant vector.

Solution: (1) **Perpendicular Velocity** Compute velocity:

$$\vec{v} = \frac{d\vec{r}}{dt} = -\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j}$$

Compute the dot product to check for perpendicularity:

$$\vec{v} \cdot \vec{r} = (-\omega \sin \omega t)(\cos \omega t) + (\omega \cos \omega t)(\sin \omega t) = 0$$

Since $\vec{v} \cdot \vec{r} = 0$, the velocity is perpendicular to the position vector ($\vec{v} \perp \vec{r}$).

(2) Acceleration Directed to Origin Compute acceleration:

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = -\omega^2 \cos \omega t \hat{i} - \omega^2 \sin \omega t \hat{j} = -\omega^2 (\cos \omega t \hat{i} + \sin \omega t \hat{j}) = -\omega^2 \vec{r}$$

The negative sign indicates the direction of acceleration is opposite to the position vector (directed towards the origin). Magnitude: $|\vec{a}| = |-\omega^2 \vec{r}| = \omega^2 |\vec{r}|$, which is proportional to the distance from the origin.

(3) Constant Cross Product Compute $\vec{r} \times \vec{v}$:

$$\vec{r} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \omega t & \sin \omega t & 0 \\ -\omega \sin \omega t & \omega \cos \omega t & 0 \end{vmatrix} = \hat{k}(\omega \cos^2 \omega t - (-\omega \sin^2 \omega t))$$

$$\vec{r} \times \vec{v} = \omega(\cos^2 \omega t + \sin^2 \omega t) \hat{k} = \omega \hat{k}$$

Since ω is constant and $\hat{k}$ is a fixed unit vector, $\vec{r} \times \vec{v} = \omega \hat{k}$ is a constant vector. (Proved)

2.5 4. Practice Problems

1. If $\vec{r} = t^2 \hat{i} - t \hat{j} + (2t + 1) \hat{k}$, find at $t = 0$, the values of:

- $\frac{d\vec{r}}{dt}$
- $\frac{d^2\vec{r}}{dt^2}$
- $\left| \frac{d\vec{r}}{dt} \right|$
- $\left| \frac{d^2\vec{r}}{dt^2} \right|$

2.6 What Comes Next

3 Lecture 03: Concept of Gradient and Geometric Meaning

Series: Vector Calculus **Lecture:** 03 of 14 **Video:** <https://www.youtube.com/watch?v=fqWikeRWBfk>

3.1 Prerequisites

We know that a scalar point function $\phi(x, y, z) = \text{Constant}$ defines a 3D **Surface**. For example, $\phi(x, y, z) = x^2 + xyz + 2 = \text{Constant}$. To study how these scalar fields change, we need a special operator that can take derivatives in all spatial directions at once.

3.2 1. The Del Operator and Gradient

3.2.1 The Del Operator (∇)

In vector calculus, we use the **Del Operator** (or Nabla), denoted by ∇ . It is a vector differential operator:

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

It functions as a vector differential operator, analogous to the scalar differential operator $\frac{d}{dx}$.

3.2.2 The Gradient ($\text{grad } \phi$)

When we apply the del operator to a scalar point function ϕ , we get the **gradient** of ϕ :

$$\nabla \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

Crucial Characteristic: The gradient of a scalar point function always results in a **vector quantity**.

3.2.3 Geometric Meaning

$\nabla \phi$ is the **normal (perpendicular) vector** to the surface defined by $\phi(x, y, z) = \text{Constant}$ at any given point (x, y, z) .

3.3 2. Properties of Gradient (With Proofs)

Let f and g be two scalar point functions, and $\vec{a}$ be a constant vector.

3.3.1 Property I: Gradient of Sum/Difference

Statement: $\nabla(f \pm g) = \nabla f \pm \nabla g$

Proof:

$$\begin{aligned} \nabla(f \pm g) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (f \pm g) \\ &= \hat{i} \left(\frac{\partial f}{\partial x} \pm \frac{\partial g}{\partial x} \right) + \hat{j} \left(\frac{\partial f}{\partial y} \pm \frac{\partial g}{\partial y} \right) + \hat{k} \left(\frac{\partial f}{\partial z} \pm \frac{\partial g}{\partial z} \right) \\ &= \left(\hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \right) \pm \left(\hat{i} \frac{\partial g}{\partial x} + \hat{j} \frac{\partial g}{\partial y} + \hat{k} \frac{\partial g}{\partial z} \right) = \nabla f \pm \nabla g \quad (\text{Proved}) \end{aligned}$$

3.3.2 Property II: Gradient of Product

Statement: $\nabla(fg) = f\nabla g + g\nabla f$

Proof:

$$\begin{aligned}\nabla(fg) &= \hat{i} \frac{\partial}{\partial x}(fg) + \hat{j} \frac{\partial}{\partial y}(fg) + \hat{k} \frac{\partial}{\partial z}(fg) \\ &= \hat{i} \left[f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right] + \hat{j} \left[f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right] + \hat{k} \left[f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right] \\ &= f \left[\hat{i} \frac{\partial g}{\partial x} + \hat{j} \frac{\partial g}{\partial y} + \hat{k} \frac{\partial g}{\partial z} \right] + g \left[\hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \right] = f\nabla g + g\nabla f \quad (\text{Proved})\end{aligned}$$

3.3.3 Property III: Gradient of Quotient

Statement: $\nabla \left(\frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}, \quad g \neq 0$

Proof:

$$\begin{aligned}\nabla \left(\frac{f}{g} \right) &= \hat{i} \frac{\partial}{\partial x} \left(\frac{f}{g} \right) + \hat{j} \frac{\partial}{\partial y} \left(\frac{f}{g} \right) + \hat{k} \frac{\partial}{\partial z} \left(\frac{f}{g} \right) \\ &= \hat{i} \left[\frac{g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x}}{g^2} \right] + \hat{j} \left[\frac{g \frac{\partial f}{\partial y} - f \frac{\partial g}{\partial y}}{g^2} \right] + \hat{k} \left[\frac{g \frac{\partial f}{\partial z} - f \frac{\partial g}{\partial z}}{g^2} \right] \\ &= \frac{g \left[\hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \right] - f \left[\hat{i} \frac{\partial g}{\partial x} + \hat{j} \frac{\partial g}{\partial y} + \hat{k} \frac{\partial g}{\partial z} \right]}{g^2} = \frac{g\nabla f - f\nabla g}{g^2} \quad (\text{Proved})\end{aligned}$$

3.3.4 Property IV: Operator Action on Position Vector

Statement: $(\vec{a} \cdot \nabla) \vec{r} = \vec{a}$

Proof: Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and constant vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$. Expand the operator:

$$\vec{a} \cdot \nabla = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) = a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z}$$

Apply to $\vec{r}$:

$$\begin{aligned}(\vec{a} \cdot \nabla) \vec{r} &= \left(a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z} \right) (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= a_1(\hat{i}) + a_2(\hat{j}) + a_3(\hat{k}) = \vec{a} \quad (\text{Proved})\end{aligned}$$

3.4 3. Solved Problems

3.4.1 Problem 1: Basic Gradient Calculation

Question: If $f = x^3 + y^3 + 3xy$, find $\text{grad } f$.

Solution:

$$\begin{aligned}\text{grad } f &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^3 + y^3 + 3xy) \\ \text{grad } f &= (3x^2 + 3y)\hat{i} + (3y^2 + 3x)\hat{j} + 0\hat{k} \\ \text{grad } f &= 3(x^2 + y)\hat{i} + 3(y^2 + x)\hat{j}\end{aligned}$$

3.4.2 Problem 2: Gradient Evaluation at a Point

Question: If $\phi(x, y, z) = 2x^2y^3 - 3y^2z^3$, find $\nabla\phi$ at the point $(1, -1, 1)$.

Solution: Calculate partial derivatives:

$$\frac{\partial\phi}{\partial x} = 4xy^3, \quad \frac{\partial\phi}{\partial y} = 6x^2y^2 - 6yz^3, \quad \frac{\partial\phi}{\partial z} = -9y^2z^2$$

Combine to form gradient $\nabla\phi$:

$$\nabla\phi = 4xy^3\hat{i} + (6x^2y^2 - 6yz^3)\hat{j} - 9y^2z^2\hat{k}$$

Evaluate at point $(1, -1, 1)$:

$$\begin{aligned} \left. \frac{\partial\phi}{\partial x} \right|_{(1, -1, 1)} &= 4(1)(-1)^3 = -4 \\ \left. \frac{\partial\phi}{\partial y} \right|_{(1, -1, 1)} &= 6(1)^2(-1)^2 - 6(-1)(1)^3 = 6 + 6 = 12 \\ \left. \frac{\partial\phi}{\partial z} \right|_{(1, -1, 1)} &= -9(-1)^2(1)^2 = -9 \end{aligned}$$

Therefore:

$$\nabla\phi \Big|_{(1, -1, 1)} = -4\hat{i} + 12\hat{j} - 9\hat{k}$$

3.4.3 Problem 3: Gradient of a Scalar Triple Product

Question: If $\vec{a}$ and $\vec{b}$ are constant vectors, prove that $\nabla[\vec{r} \cdot \vec{a} \times \vec{b}] = \vec{a} \times \vec{b}$.

Solution: Let the cross product $\vec{A} = \vec{a} \times \vec{b}$. Since $\vec{a}, \vec{b}$ are constant, $\vec{A}$ is a constant vector. The scalar triple product is a dot product:

$$[\vec{r} \cdot \vec{a} \times \vec{b}] = \vec{r} \cdot (\vec{a} \times \vec{b}) = \vec{r} \cdot \vec{A}$$

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$.

$$\vec{r} \cdot \vec{A} = A_1x + A_2y + A_3z$$

Differentiating:

$$\begin{aligned} \nabla(\vec{r} \cdot \vec{A}) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (A_1x + A_2y + A_3z) \\ &= A_1\hat{i} + A_2\hat{j} + A_3\hat{k} = \vec{A} = \vec{a} \times \vec{b} \quad (\text{Proved}) \end{aligned}$$

3.4.4 Problem 4: Gradient of r^n

Question: If $r^2 = x^2 + y^2 + z^2$, find $\text{grad } r^n$.

Solution (Chain Rule Method): Differentiating $r^2 = x^2 + y^2 + z^2$ with respect to x :

$$2r \frac{\partial r}{\partial x} = 2x \implies \frac{\partial r}{\partial x} = \frac{x}{r}$$

Now apply the chain rule:

$$\begin{aligned} \nabla r^n &= \hat{i} \frac{\partial(r^n)}{\partial x} + \hat{j} \frac{\partial(r^n)}{\partial y} + \hat{k} \frac{\partial(r^n)}{\partial z} = \sum \hat{i} \left(nr^{n-1} \frac{\partial r}{\partial x} \right) \\ \nabla r^n &= \sum \hat{i} \left(nr^{n-1} \frac{x}{r} \right) = nr^{n-2} \sum x\hat{i} = nr^{n-2}(x\hat{i} + y\hat{j} + z\hat{k}) = nr^{n-2}\vec{r} \end{aligned}$$

3.4.5 Problem 5: Cross Product of Gradient of $f(r)$ and Position Vector

Question: If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $|\vec{r}| = r$, prove that $\nabla f(r) \times \vec{r} = 0$.

Solution: Differentiate $f(r)$ using the chain rule:

$$\begin{aligned}\nabla f(r) &= \sum \hat{i} \frac{\partial f(r)}{\partial x} = \sum \hat{i} f'(r) \frac{\partial r}{\partial x} = \sum \hat{i} f'(r) \frac{x}{r} \\ \nabla f(r) &= \frac{f'(r)}{r} (x\hat{i} + y\hat{j} + z\hat{k}) = \frac{f'(r)}{r} \vec{r}\end{aligned}$$

Take the cross product with $\vec{r}$:

$$\nabla f(r) \times \vec{r} = \left(\frac{f'(r)}{r} \vec{r} \right) \times \vec{r} = \frac{f'(r)}{r} (\vec{r} \times \vec{r})$$

Since the cross product of any vector with itself is zero:

$$\nabla f(r) \times \vec{r} = 0 \quad (\text{Proved})$$

3.5 4. Homework Problems

1. If $\phi = 3x^2y - y^3z^2$, find $\text{grad } \phi$ at the point $(1, -2, -1)$. **Answer:** $-12\hat{i} - 9\hat{j} - 16\hat{k}$
2. If $f = x^3 + y^3 + 3xyz$, find $\text{grad } f$. **Answer:** $3(x^2 + yz)\hat{i} + 3(y^2 + zx)\hat{j} + 3(z^2 + xy)\hat{k}$
3. If $\vec{a} = 2x^2\hat{i} - 3yz\hat{j} + xz^2\hat{k}$ and $\phi = 2z^2 - x^3y$, find $\vec{a} \times \nabla \phi$ at the point $(1, -1, 1)$. **Answer:** $13\hat{i} - 5\hat{j} - 11\hat{k}$
4. If $r^2 = x^2 + y^2 + z^2$, find $\text{grad } r$. **Answer:** $\hat{r}$ (or $\frac{\vec{r}}{r}$)
5. Evaluate $\text{grad } e^{r^2}$, where $r^2 = x^2 + y^2 + z^2$. **Answer:** $2e^{r^2} \vec{r}$

3.6 What Comes Next

4 Lecture 04: Concept of Directional Derivative

Series: Vector Calculus Lecture: 04 of 14 Video: https://www.youtube.com/watch?v=UcEozy_ig_U

4.1 Prerequisites

We previously learned that the gradient of a scalar point function ϕ , denoted by $\nabla\phi$, gives a vector that is perfectly normal (perpendicular) to the surface. But what if we want to know how the surface changes in an arbitrary direction, not just the normal direction?

4.2 1. Directional Derivative

The **Directional Derivative (D.D.)** is the rate of change of a scalar surface $\phi(x, y, z)$ at a specific point $P(x_1, y_1, z_1)$ along the direction of a given vector $\vec{a}$.

To calculate it, we evaluate the gradient of ϕ at the given point, and then project it onto the desired direction by taking the dot product with the unit vector $\hat{a}$.

$$\text{D.D.} = \frac{d\phi}{ds} = \hat{a} \cdot \nabla\phi$$

Where: - $\nabla\phi$ is the gradient vector evaluated at point P . - $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$ is the unit vector in the direction of vector $\vec{a}$.

Because the dot product yields a scalar, the directional derivative is a scalar magnitude.

4.2.1 Maximum Directional Derivative

The dot product $\hat{a} \cdot \nabla\phi$ equals $|\nabla\phi| \cos\theta$. This value is maximized when $\cos\theta = 1$, which occurs when $\theta = 0$. This means the directional derivative is **maximum in the exact direction of the normal vector** ($\nabla\phi$). The maximum possible magnitude of the directional derivative is simply the magnitude of the gradient:

$$\text{Max D.D.} = |\nabla\phi|$$

4.2.2 Unit Normal Vector

Since the gradient $\nabla\phi$ provides a vector normal to the surface, we can easily find the **unit normal vector** $\hat{n}$ by dividing the gradient by its own magnitude:

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

4.3 2. Solved Problems

4.3.1 Problem 1: Basic Directional Derivative

Question: Find the directional derivative of $\phi = (x^2 + y^2 + z^2)^{-1/2}$ at the point $P(3, 1, 2)$ in the direction of the vector $\vec{a} = y\hat{i} + zx\hat{j} + xy\hat{k}$.

Solution: First, compute the gradient $\nabla\phi$:

$$\frac{\partial\phi}{\partial x} = -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2x) = -x(x^2 + y^2 + z^2)^{-3/2}$$

By symmetry, we get similar terms for y and z .

$$\nabla\phi = -(x^2 + y^2 + z^2)^{-3/2}(x\hat{i} + y\hat{j} + z\hat{k})$$

Evaluate the gradient at $P(3, 1, 2)$:

$$x^2 + y^2 + z^2 = 3^2 + 1^2 + 2^2 = 14$$

$$\nabla\phi|_{(3,1,2)} = \frac{-(3\hat{i} + \hat{j} + 2\hat{k})}{14\sqrt{14}}$$

Next, find the unit vector $\hat{a}$ along $\vec{a}$ at the point $P(3, 1, 2)$:

$$\vec{a}|_{(3,1,2)} = (1)(2)\hat{i} + (2)(3)\hat{j} + (3)(1)\hat{k} = 2\hat{i} + 6\hat{j} + 3\hat{k}$$

$$|\vec{a}| = \sqrt{2^2 + 6^2 + 3^2} = \sqrt{49} = 7 \implies \hat{a} = \frac{2\hat{i} + 6\hat{j} + 3\hat{k}}{7}$$

Finally, take the dot product $\hat{a} \cdot \nabla\phi$:

$$\begin{aligned} \text{D.D.} &= \left(\frac{2\hat{i} + 6\hat{j} + 3\hat{k}}{7} \right) \cdot \left(\frac{-(3\hat{i} + \hat{j} + 2\hat{k})}{14\sqrt{14}} \right) \\ \text{D.D.} &= \frac{-[2(3) + 6(1) + 3(2)]}{98\sqrt{14}} = \frac{-18}{98\sqrt{14}} = -\frac{9}{49\sqrt{14}} \end{aligned}$$

4.3.2 Problem 2: Direction Defined by Points

Question: Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ , where Q is the point $(5, 0, 4)$.

Solution: First, compute the gradient and evaluate it at $P(1, 2, 3)$:

$$\nabla\phi = 2x\hat{i} - 2y\hat{j} + 4z\hat{k}$$

$$\nabla\phi|_{(1,2,3)} = 2(1)\hat{i} - 2(2)\hat{j} + 4(3)\hat{k} = 2\hat{i} - 4\hat{j} + 12\hat{k}$$

Next, construct the direction vector $\vec{PQ}$:

$$\vec{PQ} = \vec{OQ} - \vec{OP} = (5\hat{i} + 0\hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = 4\hat{i} - 2\hat{j} + \hat{k}$$

Find the unit vector $\hat{a}$:

$$|\vec{PQ}| = \sqrt{16 + 4 + 1} = \sqrt{21} \implies \hat{a} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}}$$

Compute the directional derivative $\hat{a} \cdot \nabla\phi$:

$$\text{D.D.} = \left(\frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}} \right) \cdot (2\hat{i} - 4\hat{j} + 12\hat{k}) = \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}}$$

4.3.3 Problem 3: Finding Unknown Constants

Question: Find the constants a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at the point $(1, 2, -1)$ has a maximum magnitude of 64 in the direction parallel to the Z -axis.

Solution: First, compute the gradient and evaluate at $(1, 2, -1)$:

$$\nabla\phi = (ay^2 + 3cz^2x^2)\hat{i} + (2axy + bz)\hat{j} + (by + 2czx^3)\hat{k}$$

$$\nabla\phi|_{(1,2,-1)} = (4a + 3c)\hat{i} + (4a - b)\hat{j} + (2b - 2c)\hat{k}$$

The directional derivative is maximum in the direction of the gradient. For this maximum to occur parallel to the Z -axis (the k direction), the $\hat{i}$ and $\hat{j}$ components must be zero: 1. $4a + 3c = 0 \implies c = -\frac{4}{3}a$ 2. $4a - b = 0 \implies b = 4a$

Also, the magnitude of the gradient must equal the maximum magnitude of 64. Since only the $\hat{k}$ component survives, it must be positive and equal to 64:

$$2b - 2c = 64 \implies b - c = 32$$

Substitute b and c in terms of a :

$$4a - \left(-\frac{4}{3}a\right) = 32 \implies \frac{16}{3}a = 32 \implies a = 6$$

Now, substitute $a = 6$ back to find b and c :

$$b = 4(6) = 24$$

$$c = -\frac{4}{3}(6) = -8$$

Final Answer: $a = 6, b = 24, c = -8$.

4.3.4 Problem 4: Angle Between Surfaces

Question: Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.

Solution: The angle between two surfaces is the angle between their normal vectors. Let's find the gradient of both surfaces.

$$\phi_1 = x^2 + y^2 + z^2 - 9 \implies \nabla\phi_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\phi_2 = x^2 + y^2 - z - 3 \implies \nabla\phi_2 = 2x\hat{i} + 2y\hat{j} - \hat{k}$$

Evaluate both gradients at $(2, -1, 2)$ to find the normal vectors $\vec{n}_1$ and $\vec{n}_2$:

$$\vec{n}_1 = 4\hat{i} - 2\hat{j} + 4\hat{k} \implies |\vec{n}_1| = \sqrt{16 + 4 + 16} = 6$$

$$\vec{n}_2 = 4\hat{i} - 2\hat{j} - \hat{k} \implies |\vec{n}_2| = \sqrt{16 + 4 + 1} = \sqrt{21}$$

Use the dot product formula $\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$ to find the angle:

$$\vec{n}_1 \cdot \vec{n}_2 = (4)(4) + (-2)(-2) + (4)(-1) = 16 + 4 - 4 = 16$$

$$\theta = \cos^{-1}\left(\frac{16}{6\sqrt{21}}\right) = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

4.3.5 Problem 5: Combined Concepts

Question: For the surface $\phi = x^2yz + 4xz^2$ at the point $(1, -2, -1)$: 1. Find the directional derivative in the direction $2\hat{i} - \hat{j} - 2\hat{k}$. 2. In what direction is the directional derivative maximum, and what is its magnitude? 3. Find the unit normal vector to the surface.

Solution: First, evaluate the gradient at $(1, -2, -1)$:

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}$$

$$\nabla\phi|_{(1,-2,-1)} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Part 1: Directional Derivative Find unit vector $\hat{a}$ for $2\hat{i} - \hat{j} - 2\hat{k}$:

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$$

$$\text{D.D.} = \hat{a} \cdot \nabla\phi = \left(\frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}\right) \cdot (8\hat{i} - \hat{j} - 10\hat{k}) = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

Part 2: Maximum Directional Derivative The maximum directional derivative is in the direction of the gradient itself: $8\hat{i} - \hat{j} - 10\hat{k}$. Its magnitude is $|\nabla\phi|$:

$$\text{Max D.D.} = \sqrt{8^2 + (-1)^2 + (-10)^2} = \sqrt{64 + 1 + 100} = \sqrt{165}$$

Part 3: Unit Normal Vector Divide the gradient by its magnitude to find the unit normal vector:

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{8\hat{i} - \hat{j} - 10\hat{k}}{\sqrt{165}}$$

4.4 3. Homework Problems

1. In what direction from the point $(2, 1, -1)$ is the directional derivative of $\phi = x^2yz^3$ maximum, and what is its magnitude?
 - **Answer:** Maximum in direction of $-4\hat{i} - 4\hat{j} + 12\hat{k}$ with magnitude $4\sqrt{11}$.
2. Find the maximum value of the directional derivative of $\phi = 2x^2 + 3y^2 + 5z^2$ at the point $(1, 1, -4)$.
 - **Answer:** $2\sqrt{413}$.
3. Find the directional derivative of $\phi = x^2 - 2y^2 + 4z^2$ at the point $(1, 1, -1)$ in the direction $2\hat{i} + \hat{j} - \hat{k}$. In what direction is the directional derivative of ϕ at $(1, 1, -1)$ maximum, and what is its value?
 - **Answer:** Directional Derivative: $\frac{4\sqrt{6}}{3}$. Maximum direction: $2\hat{i} - 4\hat{j} - 8\hat{k}$. Maximum value: $2\sqrt{21}$.

4.5 What Comes Next

We have seen how the del operator ∇ applies to a scalar point function to produce a gradient vector. Next, we will see what happens when we apply the del operator to a vector point function. This introduces the concepts of **Divergence** (dot product) and **Curl** (cross product).

5 Lecture 05: Concept of Curl and Divergence

Series: Vector Calculus **Lecture:** 05 of 14 **Video:** <https://www.youtube.com/watch?v=Zdd8Tf5tkvM>

5.1 Prerequisites

We have seen that applying the Del operator (∇) to a scalar function produces the gradient (a vector). In this lecture, we apply the Del operator to a **vector point function** $\vec{F}(x, y, z) = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$. Because it is a vector operator acting on a vector function, we can do this in two ways: via a dot product (Divergence) or a cross product (Curl).

5.2 1. Divergence of a Vector

The **divergence** of a vector field $\vec{F}$ is defined as the dot product of the Del operator and $\vec{F}$. It is written as $\text{div } \vec{F}$ or $\nabla \cdot \vec{F}$.

$$\begin{aligned}\text{div } \vec{F} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (F_1\hat{i} + F_2\hat{j} + F_3\hat{k}) \\ \text{div } \vec{F} &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\end{aligned}$$

Because it is a dot product, the divergence is a **scalar quantity**. Physically, it represents the net flux per unit volume emitting from a point (e.g., heat radiating from a source, or fluid expanding outward).

5.2.1 Solenoidal Vectors

If the divergence of a vector field is zero everywhere ($\nabla \cdot \vec{F} = 0$), the vector is called **solenoidal**. This means there is no net expansion or compression. It behaves like an incompressible fluid with no sources or sinks.

5.3 2. Curl of a Vector

The **curl** of a vector field $\vec{F}$ is defined as the cross product of the Del operator and $\vec{F}$. It is written as $\text{curl } \vec{F}$ or $\nabla \times \vec{F}$. We compute this using the standard cross product determinant:

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Expanding this gives:

$$\text{curl } \vec{F} = \hat{i} \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) + \hat{j} \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) + \hat{k} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

Because it is a cross product, the curl is a **vector quantity**. Physically, it measures the rotational tendency or “spin” of the vector field at a point (like a vortex or whirlpool).

5.3.1 Irrotational Vectors

If the curl of a vector field is the zero vector everywhere, i.e., $\nabla \times \vec{F} = \vec{0}$, the vector is called **irrotational**. This means the field has absolutely no rotational flow or twisting motion.

5.4 3. Important Vector Identities

5.4.1 Identity 1: Divergence of Curl is Zero

The divergence of the curl of any vector field is always exactly zero:

$$\text{div } (\text{curl } \vec{F}) = \nabla \cdot (\nabla \times \vec{F}) = 0$$

Proof Outline: Evaluating $\nabla \cdot (\nabla \times \vec{F})$ produces mixed partial derivatives like $\frac{\partial^2 F_3}{\partial x \partial y} - \frac{\partial^2 F_3}{\partial y \partial x}$. Assuming continuity, these mixed partials are equal and cancel out in pairs, resulting in exactly 0.

5.4.2 Identity 2: Curl of Curl

The curl of the curl is related to the gradient of divergence and the Laplacian:

$$\text{curl}(\text{curl } \vec{F}) = \text{grad}(\text{div } \vec{F}) - \nabla^2 \vec{F}$$

5.5 4. Solved Problems

5.5.1 Problem 1: Solenoidal Condition

Question: Determine a such that the vector $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal.

Solution: Since the vector is solenoidal, $\nabla \cdot \vec{F} = 0$.

$$\begin{aligned} \frac{\partial}{\partial x}(x + 3y) + \frac{\partial}{\partial y}(y - 2z) + \frac{\partial}{\partial z}(x + az) &= 0 \\ 1 + 1 + a &= 0 \implies a = -2 \end{aligned}$$

5.5.2 Problem 2: Divergence at a Point

Question: If $\vec{F} = xy \sin z \hat{i} + y^2 \sin x \hat{j} + z^2 \sin xy \hat{k}$, find $\text{div } \vec{F}$ at $(0, \pi/2, \pi/2)$.

Solution:

$$\begin{aligned} \text{div } \vec{F} &= \frac{\partial}{\partial x}(xy \sin z) + \frac{\partial}{\partial y}(y^2 \sin x) + \frac{\partial}{\partial z}(z^2 \sin xy) \\ \text{div } \vec{F} &= y \sin z + 2y \sin x + 2z \sin xy \end{aligned}$$

Evaluate at $x = 0, y = \pi/2, z = \pi/2$:

$$\text{div } \vec{F} = \left(\frac{\pi}{2}\right) \sin\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2}\right) \sin(0) + 2\left(\frac{\pi}{2}\right) \sin(0) = \frac{\pi}{2}(1) + 0 + 0 = \frac{\pi}{2}$$

5.5.3 Problem 3: Properties of Position Vector

Question: If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, prove that $\text{div } \vec{r} = 3$ and $\text{curl } \vec{r} = \vec{0}$.

Solution: For divergence:

$$\nabla \cdot \vec{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

For curl:

$$\nabla \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \hat{i}(0 - 0) + \hat{j}(0 - 0) + \hat{k}(0 - 0) = \vec{0}$$

5.5.4 Problem 4: Finding Divergence and Curl

Question: Find the divergence and curl of $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j} + (y^2 - xy)\hat{k}$.

Solution: Divergence:

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(x^2 - y^2) + \frac{\partial}{\partial y}(2xy) + \frac{\partial}{\partial z}(y^2 - xy) = 2x + 2x + 0 = 4x$$

Curl:

$$\begin{aligned} \text{curl } \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 & 2xy & y^2 - xy \end{vmatrix} \\ &= \hat{i}(2y - 0) + \hat{j}(0 - (-y)) + \hat{k}(2y - (-2y)) = (2y - x)\hat{i} + y\hat{j} + 4y\hat{k} \end{aligned}$$

5.5.5 Problem 5: Curl of Exponential Field

Question: If $\vec{V} = e^{xyz}(\hat{i} + \hat{j} + \hat{k})$, find $\text{curl } \vec{V}$.

Solution:

$$\text{curl } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xyz} & e^{xyz} & e^{xyz} \end{vmatrix}$$

Evaluate the $\hat{i}$ component:

$$\frac{\partial}{\partial y}(e^{xyz}) - \frac{\partial}{\partial z}(e^{xyz}) = xze^{xyz} - xye^{xyz} = x(z - y)e^{xyz}$$

Similarly for $\hat{j}$ and $\hat{k}$:

$$\text{curl } \vec{V} = e^{xyz}[x(z - y)\hat{i} + y(x - z)\hat{j} + z(y - x)\hat{k}]$$

5.5.6 Problem 6: Evaluating Irrotational Flow

Question: Show that $\vec{F} = (\sin y + z)\hat{i} + (x \cos y - z)\hat{j} + (x - y)\hat{k}$ is irrotational.

Solution: Compute the curl:

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin y + z & x \cos y - z & x - y \end{vmatrix} \\ &= \hat{i}(-1 - (-1)) + \hat{j}(1 - 1) + \hat{k}(\cos y - \cos y) = \vec{0} \end{aligned}$$

Since $\nabla \times \vec{F} = \vec{0}$, the vector is irrotational.

5.5.7 Problem 7: Curl of a Cross Product Identity

Question: If $\vec{a}$ is a constant vector and $\vec{r}$ is the position vector, prove that $\text{curl } (\vec{r} \times \vec{a}) = -2\vec{a}$.

Solution: Use the vector identity $\nabla \times (\vec{A} \times \vec{B}) = \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B}$. Let $\vec{A} = \vec{r}$ and $\vec{B} = \vec{a}$. Since $\vec{a}$ is constant, $\nabla \cdot \vec{a} = 0$ and $(\vec{r} \cdot \nabla)\vec{a} = \vec{0}$. From earlier, $\nabla \cdot \vec{r} = 3$ and $(\vec{a} \cdot \nabla)\vec{r} = \vec{a}$. Substitute these into the identity:

$$\text{curl } (\vec{r} \times \vec{a}) = \vec{r}(0) - \vec{a}(3) + \vec{a} - \vec{0} = -3\vec{a} + \vec{a} = -2\vec{a} \quad (\text{Proved})$$

5.5.8 Problem 8: Operations on $r^n \vec{r}$

Question: Prove that $\text{div}(r^n \vec{r}) = (n + 3)r^n$ and $\text{curl}(r^n \vec{r}) = \vec{0}$. For what value of n is it solenoidal?

Solution: First, note $\frac{\partial r}{\partial x} = \frac{x}{r}$, similarly for y and z . **Divergence:**

$$\nabla \cdot (r^n x \hat{i} + r^n y \hat{j} + r^n z \hat{k}) = \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z)$$

Apply product rule:

$$\frac{\partial}{\partial x}(r^n x) = r^n + x \left(n r^{n-1} \frac{x}{r} \right) = r^n + n x^2 r^{n-2}$$

Summing x, y, z :

$$\text{div}(r^n \vec{r}) = 3r^n + n r^{n-2}(x^2 + y^2 + z^2) = 3r^n + n r^{n-2}(r^2) = (n + 3)r^n$$

For solenoidal, $\text{div} = 0 \implies (n + 3)r^n = 0 \implies n = -3$.

Curl:

$$\text{curl}(r^n \vec{r}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^n x & r^n y & r^n z \end{vmatrix}$$

$\hat{i}$ component: $\frac{\partial}{\partial y}(r^n z) - \frac{\partial}{\partial z}(r^n y) = z(n r^{n-1} \frac{y}{r}) - y(n r^{n-1} \frac{z}{r}) = n y z r^{n-2} - n y z r^{n-2} = 0$. By symmetry, all components are 0, so $\text{curl}(r^n \vec{r}) = \vec{0}$.

5.6 What Comes Next

We have completed our study of vector differential operators (Gradient, Divergence, and Curl). The next major topic in vector calculus is **Vector Integration**, starting with Line Integrals and calculating work done by a force.

6 Lecture 06: Concept of Vector Integration and Line Integration

Series: Vector Calculus **Lecture:** 06 of 14 **Video:** <https://www.youtube.com/watch?v=0CRFfOOSA9c>

6.1 Prerequisites

We have thoroughly covered vector differentiation (Gradient, Divergence, and Curl). Now we move in the opposite direction: **Vector Integration**. There are three main types of integrals in vector calculus: 1. **Line Integral** 2. **Surface Integral** 3. **Volume Integral**

A solid understanding of these integrals forms the foundation for three major theorems: Green's Theorem, Stokes' Theorem, and Gauss Divergence Theorem. We will start with standard vector integration and Line Integrals.

6.2 1. Integration of Vector Functions

Just like scalar functions, a vector point function $\vec{F}(t) = F_1(t)\hat{i} + F_2(t)\hat{j} + F_3(t)\hat{k}$ depending on a single scalar variable t can be integrated.

6.2.1 Indefinite Integral

To integrate indefinitely, integrate each component separately and add an arbitrary constant vector $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$.

$$\int \vec{F}(t)dt = \left(\int F_1(t)dt\right)\hat{i} + \left(\int F_2(t)dt\right)\hat{j} + \left(\int F_3(t)dt\right)\hat{k} + \vec{c}$$

6.2.2 Definite Integral

To integrate over an interval $[a, b]$, compute the integral and apply the upper and lower limits to each component.

$$\int_a^b \vec{F}(t)dt = \left[\vec{f}(t)\right]_a^b = \vec{f}(b) - \vec{f}(a)$$

6.3 2. Solved Problems (Vector Integration)

6.3.1 Problem 1: Basic Definite Integral

Question: If $\vec{F}(t) = (t - t^2)\hat{i} + 2t^3\hat{j} - 3\hat{k}$, find $\int_1^2 \vec{F}(t)dt$.

Solution: Integrate each component with respect to t :

$$\int_1^2 \left[(t - t^2)\hat{i} + 2t^3\hat{j} - 3\hat{k}\right] dt = \left[\left(\frac{t^2}{2} - \frac{t^3}{3}\right)\hat{i} + \frac{2t^4}{4}\hat{j} - 3t\hat{k}\right]_1^2$$

Apply the upper limit ($t = 2$):

$$\left(\frac{4}{2} - \frac{8}{3}\right)\hat{i} + \frac{16}{2}\hat{j} - 6\hat{k} = -\frac{2}{3}\hat{i} + 8\hat{j} - 6\hat{k}$$

Apply the lower limit ($t = 1$):

$$\left(\frac{1}{2} - \frac{1}{3}\right)\hat{i} + \frac{1}{2}\hat{j} - 3\hat{k} = \frac{1}{6}\hat{i} + \frac{1}{2}\hat{j} - 3\hat{k}$$

Subtract the two:

$$\left(-\frac{2}{3} - \frac{1}{6}\right)\hat{i} + \left(8 - \frac{1}{2}\right)\hat{j} + (-6 - (-3))\hat{k} = -\frac{5}{6}\hat{i} + \frac{15}{2}\hat{j} - 3\hat{k}$$

6.3.2 Problem 2: Finding Velocity and Displacement

Question: The acceleration of a particle is $\vec{a} = 12 \cos 2t\hat{i} - 8 \sin 2t\hat{j} + 16t\hat{k}$. Find velocity $\vec{v}$ and displacement $\vec{r}$ given that at $t = 0$, $\vec{v} = 0$ and $\vec{r} = 0$.

Solution: Integrate acceleration to find velocity $\vec{v}(t)$:

$$\begin{aligned}\vec{v}(t) &= \int \vec{a}(t) dt = \int (12 \cos 2t\hat{i} - 8 \sin 2t\hat{j} + 16t\hat{k}) dt \\ \vec{v}(t) &= 6 \sin 2t\hat{i} + 4 \cos 2t\hat{j} + 8t^2\hat{k} + \vec{c}_1\end{aligned}$$

Apply boundary condition $\vec{v}(0) = 0$:

$$\begin{aligned}0 &= 0\hat{i} + 4\hat{j} + 0\hat{k} + \vec{c}_1 \implies \vec{c}_1 = -4\hat{j} \\ \vec{v}(t) &= 6 \sin 2t\hat{i} + (4 \cos 2t - 4)\hat{j} + 8t^2\hat{k}\end{aligned}$$

Integrate velocity to find displacement $\vec{r}(t)$:

$$\begin{aligned}\vec{r}(t) &= \int \vec{v}(t) dt = \int [6 \sin 2t\hat{i} + (4 \cos 2t - 4)\hat{j} + 8t^2\hat{k}] dt \\ \vec{r}(t) &= -3 \cos 2t\hat{i} + (2 \sin 2t - 4t)\hat{j} + \frac{8}{3}t^3\hat{k} + \vec{c}_2\end{aligned}$$

Apply boundary condition $\vec{r}(0) = 0$:

$$\begin{aligned}0 &= -3\hat{i} + 0\hat{j} + 0\hat{k} + \vec{c}_2 \implies \vec{c}_2 = 3\hat{i} \\ \vec{r}(t) &= (3 - 3 \cos 2t)\hat{i} + (2 \sin 2t - 4t)\hat{j} + \frac{8}{3}t^3\hat{k}\end{aligned}$$

6.3.3 Problem 3: Integration Involving Dot Product

Question: Given $\vec{r}(t) = 2\hat{i} - \hat{j} + 2\hat{k}$ at $t = 2$ and $\vec{r}(t) = 4\hat{i} - 2\hat{j} + 3\hat{k}$ at $t = 3$, prove that $\int_2^3 (\vec{r} \cdot \frac{d\vec{r}}{dt}) dt = 10$.

Solution: Calculate the magnitudes of $\vec{r}$ at the boundaries: At $t = 2$: $r_1 = |\vec{r}| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$
At $t = 3$: $r_2 = |\vec{r}| = \sqrt{4^2 + (-2)^2 + 3^2} = \sqrt{29}$ Using the identity $\vec{r} \cdot \frac{d\vec{r}}{dt} = \frac{1}{2} \frac{d}{dt}(r^2)$:

$$\begin{aligned}\int_2^3 \left(\vec{r} \cdot \frac{d\vec{r}}{dt} \right) dt &= \frac{1}{2} \int_2^3 \frac{d}{dt}(r^2) dt = \frac{1}{2} [r^2]_{t=2}^{t=3} = \frac{1}{2}(r_2^2 - r_1^2) \\ &= \frac{1}{2}(29 - 9) = \frac{1}{2}(20) = 10 \quad (\text{Proved})\end{aligned}$$

6.4 3. Line Integration

A line integral computes the integral of a vector field $\vec{F}$ along a specific curve C in space. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of any point on curve C . The differential displacement vector is $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$. The line integral of $\vec{F}$ along curve C is defined by the dot product:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$$

6.4.1 How to Evaluate Line Integrals

You cannot integrate directly when x, y , and z are all mixed together. **The curve C provides the relationship between the variables.** The most reliable method is to use parametric equations. If the curve C is described by $x = x(t)$, $y = y(t)$, and $z = z(t)$ as t varies from t_1 to t_2 :

$$\int_C \vec{F} \cdot d\vec{r} = \int_{t_1}^{t_2} \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt$$

If C is a closed curve (like a circle or rectangle), the line integral is called the **circulation** of $\vec{F}$, denoted as $\oint_C \vec{F} \cdot d\vec{r}$. By standard convention, closed paths are traversed in the anti-clockwise direction.

6.5 4. Solved Problems (Line Integration)

6.5.1 Problem 4: Line Integration on a Parametric Curve

Question: Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$ and C is $\vec{r} = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ from $t = -1$ to 1 .

Solution: From the curve equation, $x = t, y = t^2, z = t^3$. Differentiating gives $dx = dt, dy = 2tdt, dz = 3t^2dt$. Substitute x, y, z into $\vec{F}$:

$$\vec{F} = (t)(t^2)\hat{i} + (t^2)(t^3)\hat{j} + (t^3)(t)\hat{k} = t^3\hat{i} + t^5\hat{j} + t^4\hat{k}$$

The displacement vector is $d\vec{r} = (\hat{i} + 2t\hat{j} + 3t^2\hat{k})dt$. Take the dot product $\vec{F} \cdot d\vec{r}$:

$$\vec{F} \cdot d\vec{r} = (t^3)(1) + (t^5)(2t) + (t^4)(3t^2) = (t^3 + 2t^6 + 3t^6)dt = (t^3 + 5t^6)dt$$

Integrate from -1 to 1 :

$$\int_{-1}^1 (t^3 + 5t^6)dt = \left[\frac{t^4}{4} + \frac{5t^7}{7} \right]_{-1}^1 = \left(\frac{1}{4} + \frac{5}{7} \right) - \left(\frac{1}{4} - \frac{5}{7} \right) = \frac{10}{7}$$

6.5.2 Problem 5: Line Integration on a Parabola

Question: Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = x^2y^2\hat{i} + y\hat{j}$ and C is the parabola $y^2 = 4x$ from $(0, 0)$ to $(4, 4)$.

Solution: The integrand is $\vec{F} \cdot d\vec{r} = x^2y^2dx + ydy$. Using the curve equation $y^2 = 4x \implies x = \frac{y^2}{4}$, we substitute it into the dx term. Both x and y vary from 0 to 4 .

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^4 x^2(4x)dx + \int_0^4 ydy = \int_0^4 4x^3dx + \int_0^4 ydy \\ &= [x^4]_0^4 + \left[\frac{y^2}{2} \right]_0^4 = (256) + \left(\frac{16}{2} \right) = 256 + 8 = 264 \end{aligned}$$

6.5.3 Problem 6: Line Integration on an Arbitrary Path

Question: Evaluate $\int_C [yzdx + (zx + 1)dy + xydz]$ where C is any path from $(1, 0, 0)$ to $(2, 1, 4)$.

Solution: Since the integral is path-independent (it specifies “any path”), we choose the straight line connecting the two points. The equation of the line is $\frac{x-1}{2-1} = \frac{y-0}{1-0} = \frac{z-0}{4-0} = t$. Parametric equations: $x = t + 1, y = t, z = 4t \implies dx = dt, dy = dt, dz = 4dt$. At $(1, 0, 0)$, $t = 0$. At $(2, 1, 4)$, $t = 1$. Substitute these into the integrand:

$$(t)(4t)dt + ((4t)(t + 1) + 1)dt + (t + 1)(t)(4dt) = (4t^2 + 4t^2 + 4t + 1 + 4t^2 + 4t)dt = (12t^2 + 8t + 1)dt$$

Integrate from 0 to 1 :

$$\int_0^1 (12t^2 + 8t + 1)dt = [4t^3 + 4t^2 + t]_0^1 = 4(1) + 4(1) + 1 = 9$$

6.5.4 Problem 7: Circulation Around a Rectangle (1)

Question: Find the circulation of $\vec{F} = e^x \sin y \hat{i} + e^x \cos y \hat{j}$ along the rectangle with vertices $(0, 0), (1, 0), (1, \frac{\pi}{2}), (0, \frac{\pi}{2})$.

Solution: The integral is $\oint_C \vec{F} \cdot d\vec{r} = \oint_C (e^x \sin y dx + e^x \cos y dy)$. We split it into four segments: 1. **OA from $(0, 0)$ to $(1, 0)$:** $y = 0 \implies dy = 0$. x from 0 to 1 . $\int_{OA} = \int_0^1 e^x(0)dx = 0$ 2. **AB from $(1, 0)$ to $(1, \frac{\pi}{2})$:** $x = 1 \implies dx = 0$. y from 0 to $\frac{\pi}{2}$. $\int_{AB} = \int_0^{\pi/2} e^1 \cos y dy = e[\sin y]_0^{\pi/2} = e$ 3. **BC from $(1, \frac{\pi}{2})$ to $(0, \frac{\pi}{2})$:** $y = \frac{\pi}{2} \implies dy = 0$. x from 1 to 0 . $\int_{BC} = \int_1^0 e^x \sin(\frac{\pi}{2})dx = \int_1^0 e^x dx = [e^x]_1^0 = 1 - e$ 4. **CO from $(0, \frac{\pi}{2})$ to $(0, 0)$:** $x = 0 \implies dx = 0$. y from $\frac{\pi}{2}$ to 0 . $\int_{CO} = \int_{\pi/2}^0 e^0 \cos y dy = [\sin y]_{\pi/2}^0 = 0 - 1 = -1$

Summing all segments:

$$\oint_C \vec{F} \cdot d\vec{r} = 0 + e + (1 - e) - 1 = 0$$

6.5.5 Problem 8: Circulation Around a Rectangle (2)

Question: Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ and C is the rectangle bounded by $y = 0, x = a, y = b, x = 0$.

Solution: The integral is $\int_C ((x^2 + y^2)dx - 2xydy)$. Splitting into four segments OA, AC, CB, BO : 1. OA ($y = 0 \implies dy = 0, x$ from 0 to a): $\int_{OA} = \int_0^a x^2 dx = \frac{a^3}{3}$ 2. AC ($x = a \implies dx = 0, y$ from 0 to b): $\int_{AC} = \int_0^b -2ay dy = -2a[\frac{y^2}{2}]_0^b = -ab^2$ 3. CB ($y = b \implies dy = 0, x$ from a to 0): $\int_{CB} = \int_a^0 (x^2 + b^2)dx = [\frac{x^3}{3} + b^2x]_a^0 = -\frac{a^3}{3} - ab^2$ 4. BO ($x = 0 \implies dx = 0, y$ from b to 0): $\int_{BO} = \int_b^0 (0)dy = 0$

Summing all segments:

$$\int_C \vec{F} \cdot d\vec{r} = \frac{a^3}{3} - ab^2 - \frac{a^3}{3} - ab^2 + 0 = -2ab^2$$

6.5.6 Problem 9: Line Integration on a Helix

Question: Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ and C is the curve $\vec{r} = (a \cos t)\hat{i} + (b \sin t)\hat{j} + (ct)\hat{k}$ from $t = 0$ to $t = \pi/2$.

Solution: The parametric equations are $x = a \cos t, y = b \sin t, z = ct$. Differentials: $dx = -a \sin t dt, dy = b \cos t dt, dz = c dt$. Substitute into $\vec{F}$:

$$\vec{F} = (bct \sin t)\hat{i} + (act \cos t)\hat{j} + (ab \sin t \cos t)\hat{k}$$

Dot product with $d\vec{r} = (-a \sin t \hat{i} + b \cos t \hat{j} + c \hat{k})dt$:

$$\begin{aligned} \vec{F} \cdot d\vec{r} &= (-abct \sin^2 t + abct \cos^2 t + abc \sin t \cos t)dt \\ &= abc[t(\cos^2 t - \sin^2 t) + \sin t \cos t]dt = abc \left(t \cos 2t + \frac{\sin 2t}{2} \right) dt \end{aligned}$$

Integrate from 0 to $\pi/2$:

$$\int_C \vec{F} \cdot d\vec{r} = abc \int_0^{\pi/2} \left(t \cos 2t + \frac{\sin 2t}{2} \right) dt$$

Using integration by parts on $\int t \cos 2t dt = \frac{t \sin 2t}{2} - \int \frac{\sin 2t}{2} dt$:

$$\int \left(t \cos 2t + \frac{\sin 2t}{2} \right) dt = \frac{t \sin 2t}{2}$$

Evaluate limits:

$$abc \left[\frac{t \sin 2t}{2} \right]_0^{\pi/2} = abc \left(\frac{\pi}{2} \frac{\sin \pi}{2} - 0 \right) = abc(0) = 0$$

6.6 Key Takeaways

- The curve C provides the critical constraints (relationship between variables) needed to evaluate a line integral.
- By substituting the curve's parametric equations into the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$, you can convert the entire integral into a single variable t .
- For line integrals over segmented paths (like a rectangle or triangle), split the integral into a sum of integrals for each segment, defining the unique bounds and constant variables for each.

6.7 What Comes Next

We have seen how to calculate the integral over a one-dimensional path (Line Integration). Next, we will expand this into two dimensions with **Surface Integration**, where we will integrate over a 2D surface in 3D space.

7 Lecture 07: Concept of Surface and Volume Integral (Part 1)

Series: Vector Calculus Lecture: 07 of 14 Video: <https://www.youtube.com/watch?v=5BLWsAa3Yr8>

7.1 Prerequisites

We have completed our study of Line Integrals, which involve integrating a vector field over a one-dimensional path (a curve). Now, we extend our integration to two dimensions (Surface Integrals) and three dimensions (Volume Integrals).

7.2 1. Surface Integration

A surface integral computes the integral of a vector field over a 3D surface S . It is equivalent to a double integral in standard calculus, as it requires two parameters to define a surface.

The surface integral of a vector point function $\mathbf{F}$ over a surface S is given by:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$$

Where: - $\mathbf{F}$ is the vector field. - $\hat{\mathbf{n}}$ is the unit normal vector pointing outward from the surface S . - dS is the infinitesimal surface area element of S .

7.2.1 Finding the Unit Normal Vector

If the surface S is represented by a scalar equation $\phi(x, y, z) = c$, then the unit normal vector $\hat{\mathbf{n}}$ is simply the normalized gradient:

$$\hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|}$$

7.2.2 Projecting the Surface

Because the surface S exists in 3D space, we cannot integrate dS directly. We must **project** the 3D surface onto one of the 2D coordinate planes (xy , yz , or zx). This allows us to establish the limits of integration for the two projected variables.

The formulas for the projected area element dS are: 1. **Projection on xy -plane:**

$$dS = \frac{dx dy}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}|}$$

2. **Projection on yz -plane:**

$$dS = \frac{dy dz}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{i}}|}$$

3. **Projection on zx -plane:**

$$dS = \frac{dz dx}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{j}}|}$$

Rule of thumb: The denominator is the dot product of $\hat{\mathbf{n}}$ and the unit vector of the axis orthogonal to the projection plane.

7.3 2. Volume Integration

Volume integration extends the concept to a three-dimensional region (or volume) V . It is represented as a triple integral:

$$\iiint_V \mathbf{F} dV$$

In Cartesian coordinates, the volume element is simply $dV = dx dy dz$. Possible forms of volume integrals include: - $\iiint_V (\nabla \cdot \mathbf{F}) dV$ - $\iiint_V (\nabla \times \mathbf{F}) dV$

7.3.1 Setting Limits

To find the bounds for volume integration: 1. Project the 3D volume onto a 2D coordinate plane (e.g., the xy -plane). The boundary of this 2D shadow provides the limits for two variables (e.g., x and y). 2. The limits for the third variable (e.g., z) are defined by the “floor” and “ceiling” surfaces of the volume along that axis.

7.4 3. Solved Problems

7.4.1 Problem 1: Surface Integral Over a Plane

Question: If $\mathbf{F} = 6z\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + y\hat{\mathbf{k}}$, evaluate $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$, where S is the portion of the plane $2x + 3y + 6z = 12$ which lies in the first octant.

Solution: Step 1: Understand the Surface S The plane $2x + 3y + 6z = 12$ in the first octant ($x, y, z > 0$) forms a triangular surface. We find its intercepts by setting two variables to 0: - x -intercept: $(6, 0, 0)$ - y -intercept: $(0, 4, 0)$ - z -intercept: $(0, 0, 2)$

Step 2: Find the Unit Normal $\hat{\mathbf{n}}$ Let $\phi = 2x + 3y + 6z - 12$. The gradient is:

$$\nabla\phi = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$$

The magnitude is $\sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7$.

$$\hat{\mathbf{n}} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$$

Step 3: Choose a Projection Plane We project the surface S onto the xy -plane. Let's call this 2D projected region S_1 . The projection formula for dS on the xy -plane is $dS = \frac{dx dy}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}|}$.

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) \cdot \hat{\mathbf{k}} = \frac{6}{7}$$

Thus,

$$dS = \frac{dx dy}{6/7} = \frac{7}{6} dx dy$$

Step 4: Establish Limits of Integration for S_1 On the xy -plane ($z = 0$), the plane equation becomes $2x + 3y = 12$. This line bounds the triangular region S_1 . - x varies from 0 to 6. - For a given x , y varies from 0 to the line $y = \frac{12-2x}{3}$.

Step 5: Compute $\mathbf{F} \cdot \hat{\mathbf{n}}$

$$\mathbf{F} \cdot \hat{\mathbf{n}} = (6z\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + y\hat{\mathbf{k}}) \cdot \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) = \frac{1}{7}(12z - 12 + 6y)$$

Because we are integrating with respect to x and y , we must eliminate z . Using the plane equation $6z = 12 - 2x - 3y$, we substitute $12z$:

$$12z = 24 - 4x - 6y$$

$$\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{1}{7}(24 - 4x - 6y - 12 + 6y) = \frac{1}{7}(12 - 4x)$$

Step 6: Evaluate the Double Integral Substitute everything into the integral:

$$\begin{aligned} \iint_{S_1} \left[\frac{1}{7}(12 - 4x) \right] \left(\frac{7}{6} dx dy \right) &= \int_0^6 \int_0^{\frac{12-2x}{3}} \frac{1}{6}(12 - 4x) dy dx \\ &= \int_0^6 \int_0^{\frac{12-2x}{3}} \left(2 - \frac{2}{3}x \right) dy dx = \frac{1}{3} \int_0^6 \int_0^{\frac{12-2x}{3}} (6 - 2x) dy dx \end{aligned}$$

Integrate with respect to y first:

$$\frac{1}{3} \int_0^6 (6 - 2x) \left[y \right]_0^{\frac{12-2x}{3}} dx = \frac{1}{3} \int_0^6 (6 - 2x) \left(\frac{12 - 2x}{3} \right) dx$$

$$= \frac{1}{9} \int_0^6 4(3-x)(6-x) dx = \frac{4}{9} \int_0^6 (18 - 9x + x^2) dx$$

Integrate with respect to x :

$$\frac{4}{9} \left[18x - \frac{9x^2}{2} + \frac{x^3}{3} \right]_0^6 = \frac{4}{9} [108 - 162 + 72] = \frac{4}{9} [18] = 8$$

Final Answer: 8

7.5 Key Takeaways

- Surface integrals require calculating a unit normal vector ($\hat{\mathbf{n}} = \nabla\phi/|\nabla\phi|$) and computing the dot product $\mathbf{F} \cdot \hat{\mathbf{n}}$.
- To evaluate the integral, you **must project the 3D surface onto a 2D plane**. This transforms the dS element and defines the integration limits.
- Once projected, ensure the entire integrand ($\mathbf{F} \cdot \hat{\mathbf{n}}$) is expressed strictly in terms of the two projected variables.

7.6 What Comes Next

8 Lecture 08: Concept of Surface and Volume Integral (Part 2)

Series: Vector Calculus Lecture: 08 of 14 Video: <https://www.youtube.com/watch?v=cHe58sjHmgg>

8.1 Prerequisites

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$$

And solved a problem by projecting the 3D surface onto the xy -plane. In this lecture, we will demonstrate that the choice of projection plane does not alter the final result, and we will introduce **Spherical Coordinates** to drastically simplify integration over spherical surfaces.

8.2 1. Solved Problems

8.2.1 Problem 1 (Revisited): Projection on the yz -plane

Question: If $\mathbf{F} = 6z\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + y\hat{\mathbf{k}}$, evaluate $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$, where S is the portion of the plane $2x + 3y + 6z = 12$ in the first octant.

Solution:

Step 1: The Unit Normal Vector We already calculated $\hat{\mathbf{n}}$:

$$\hat{\mathbf{n}} = \frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$$

Step 2: Projection on the yz -plane When projecting onto the yz -plane, the area element is:

$$dS = \frac{dy dz}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{i}}|}$$

Compute the dot product: $\hat{\mathbf{n}} \cdot \hat{\mathbf{i}} = \frac{2}{7}$. So,

$$dS = \frac{dy dz}{2/7} = \frac{7}{2} dy dz$$

Step 3: Establish Limits on the yz -plane Set $x = 0$ in the plane equation: $3y + 6z = 12 \implies y + 2z = 4$. This line forms a triangle with the y and z axes. - y varies from 0 to 4 - z varies from 0 to the line $z = 2 - \frac{y}{2}$

Step 4: Compute the Integrand We already know:

$$\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{1}{7}(12z - 12 + 6y)$$

Because we are integrating with respect to y and z , there is no x variable to eliminate. We can use this directly!

Step 5: Evaluate the Integral

$$\iint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) dS = \int_0^4 \int_0^{2-y/2} \frac{1}{7}(12z - 12 + 6y) \left(\frac{7}{2} dz dy\right)$$

Cancel the 7's and divide the terms by 2:

$$\begin{aligned} &= \int_0^4 \int_0^{2-y/2} (6z - 6 + 3y) dz dy \\ &= \int_0^4 [3z^2 - 6z + 3yz]_0^{2-y/2} dy \end{aligned}$$

$$\begin{aligned}
&= \int_0^4 \left\{ 3 \left(2 - \frac{y}{2} \right)^2 - 6 \left(2 - \frac{y}{2} \right) + 3y \left(2 - \frac{y}{2} \right) \right\} dy \\
&= \int_0^4 \left\{ 3 \left(4 + \frac{y^2}{4} - 2y \right) - 12 + 3y + 6y - \frac{3y^2}{2} \right\} dy \\
&= \int_0^4 \left(12 + \frac{3}{4}y^2 - 6y - 12 + 9y - \frac{3y^2}{2} \right) dy \\
&= \int_0^4 \left(3y - \frac{3y^2}{4} \right) dy \\
&= 3 \left[\frac{y^2}{2} \right]_0^4 - \frac{3}{4} \left[\frac{y^3}{3} \right]_0^4 = \frac{48}{2} - 16 = 24 - 16 = 8
\end{aligned}$$

The answer is identical to the xy -plane projection.

8.2.2 Problem 2: Surface Integral Over a Sphere

Question: Evaluate $\iint_S (yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}) \cdot d\mathbf{S}$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first octant. (Note: $d\mathbf{S}$ in bold implies $\hat{\mathbf{n}} dS$)

Solution: Let $f = x^2 + y^2 + z^2 - 1$. The gradient is:

$$\nabla f = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}$$

The magnitude is $|\nabla f| = \sqrt{4(x^2 + y^2 + z^2)} = \sqrt{4(1)} = 2$. Thus, the unit normal vector is:

$$\hat{\mathbf{n}} = \frac{2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}}{2} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

Compute $\mathbf{F} \cdot \hat{\mathbf{n}}$:

$$(yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}) \cdot (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) = xyz + xyz + xyz = 3xyz$$

We need to evaluate:

$$I = \iint_S 3xyz dS$$

We will solve this using two different methods to demonstrate the power of coordinate systems.

8.2.2.1 Method 1: Cartesian Projection (on xy -plane) Project the sphere onto the xy -plane. The projection is a quarter circle bounded by $x^2 + y^2 = 1$. - x varies from 0 to 1. - y varies from 0 to $\sqrt{1-x^2}$. The area element is $dS = \frac{dx dy}{|\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}|}$. Since $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = z$, we have $dS = \frac{dx dy}{z}$. Substitute into the integral:

$$I = \iint_{S_1} 3xyz \left(\frac{dx dy}{z} \right) = 3 \iint_{S_1} xy dx dy$$

Notice how beautifully the z term cancels out! Let's evaluate:

$$\begin{aligned}
I &= 3 \int_0^1 \int_0^{\sqrt{1-x^2}} xy dy dx = 3 \int_0^1 x \left[\frac{y^2}{2} \right]_0^{\sqrt{1-x^2}} dx \\
&= \frac{3}{2} \int_0^1 x(1-x^2) dx = \frac{3}{2} \int_0^1 (x - x^3) dx \\
&= \frac{3}{2} \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{3}{2} \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{3}{2} \left(\frac{1}{4} \right) = \frac{3}{8}
\end{aligned}$$

8.2.2.2 Method 2: Spherical Polar Coordinates When integrating over a spherical surface, spherical coordinates simplify the process by removing the need for planar projections entirely. For a unit sphere ($r = 1$): - $x = \sin \theta \cos \phi$ - $y = \sin \theta \sin \phi$ - $z = \cos \theta$ - The area element $dS = \sin \theta d\theta d\phi$

For the first octant, the angles θ and ϕ both range from 0 to $\pi/2$. Substitute these into $I = \iint_S 3xyz dS$:

$$I = 3 \int_0^{\pi/2} \int_0^{\pi/2} (\sin \theta \cos \phi)(\sin \theta \sin \phi)(\cos \theta)(\sin \theta d\theta d\phi)$$

Group the terms by variable:

$$I = 3 \int_0^{\pi/2} \int_0^{\pi/2} (\sin^3 \theta \cos \theta)(\sin \phi \cos \phi) d\theta d\phi$$

Because the limits are constants, we can separate the integral into the product of two single-variable integrals:

$$I = 3 \left(\int_0^{\pi/2} \sin^3 \theta \cos \theta d\theta \right) \left(\int_0^{\pi/2} \sin \phi \cos \phi d\phi \right)$$

Using u -substitution ($u = \sin \theta$, $du = \cos \theta d\theta$ and $v = \sin \phi$, $dv = \cos \phi d\phi$):

$$I = 3 \left[\frac{\sin^4 \theta}{4} \right]_0^{\pi/2} \left[\frac{\sin^2 \phi}{2} \right]_0^{\pi/2} = 3 \left(\frac{1}{4} \right) \left(\frac{1}{2} \right) = \frac{3}{8}$$

8.3 Key Takeaways

- **Projection invariance:** Projecting a 3D surface onto the xy , yz , or zx planes will always yield the same final integral value, provided the limits and dS conversions are done correctly.
- **Spherical coordinates:** When the surface S is part of a sphere, substituting x, y, z with spherical coordinates and using $dS = r^2 \sin \theta d\theta d\phi$ eliminates the need to project the surface. The entire integral simply becomes a double integral over the angles θ and ϕ .

8.4 What Comes Next

We will continue this block with Volume Integration, exploring how to compute triple integrals over 3D volumes.

9 Lecture 09: Concept of Surface and Volume Integral (Part 3)

Series: Vector Calculus Lecture: 09 of 14 Video: https://www.youtube.com/watch?v=EWpo_cqinOI

9.1 Prerequisites

In the preceding lectures, we explored surface integrals for flat planes and spherical surfaces. This lecture builds on those concepts to tackle more complex scenarios: evaluating surface integrals over a **closed multi-face surface** (like a cube) and formally calculating a **Volume Integral** over a 3D solid using triple integration.

9.2 Solved Problems

9.2.1 Problem 1: Surface Integral Over a Closed Cube

Question: Evaluate $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$, where $\mathbf{F} = 4xz\hat{\mathbf{i}} - y^2\hat{\mathbf{j}} + yz\hat{\mathbf{k}}$ and S is the closed surface of the cube bounded by the planes $x = 0$, $x = a$, $y = 0$, $y = a$, $z = 0$, $z = a$.

Solution: The closed surface S of a cube consists of 6 distinct flat faces. To evaluate the surface integral over S , we must evaluate the integral over each of the 6 faces separately and sum the results.

Since the faces lie on or parallel to the coordinate planes, we don't need to compute gradients. We can directly identify the **outward unit normal vector** $\hat{\mathbf{n}}$ by observing which direction points away from the interior of the cube. The total integral is: $I = I_1 + I_2 + I_3 + I_4 + I_5 + I_6$.

1. Front Face ($x = a$) * Normal: Outward normal points in the $+x$ direction, so $\hat{\mathbf{n}} = \hat{\mathbf{i}}$. * **Area Element:** Projection is on the yz -plane, so $dS = dy dz$. * **Integrand:** $\mathbf{F} \cdot \hat{\mathbf{n}} = 4xz$. Since $x = a$, $\mathbf{F} \cdot \hat{\mathbf{n}} = 4az$. * **Integral:**

$$I_1 = \int_0^a \int_0^a 4az dy dz = 4a \int_0^a z dz \int_0^a dy = 4a \left(\frac{a^2}{2} \right) (a) = 2a^4$$

2. Back Face ($x = 0$) * Normal: Outward normal points in the $-x$ direction, so $\hat{\mathbf{n}} = -\hat{\mathbf{i}}$. * **Area Element:** $dS = dy dz$. * **Integrand:** $\mathbf{F} \cdot \hat{\mathbf{n}} = -4xz$. Since $x = 0$, $\mathbf{F} \cdot \hat{\mathbf{n}} = 0$. * **Integral:**

$$I_2 = 0$$

3. Right Face ($y = a$) * Normal: Outward normal points in the $+y$ direction, so $\hat{\mathbf{n}} = \hat{\mathbf{j}}$. * **Area Element:** $dS = dx dz$. * **Integrand:** $\mathbf{F} \cdot \hat{\mathbf{n}} = -y^2$. Since $y = a$, $\mathbf{F} \cdot \hat{\mathbf{n}} = -a^2$. * **Integral:**

$$I_3 = \int_0^a \int_0^a -a^2 dx dz = -a^2(a)(a) = -a^4$$

4. Left Face ($y = 0$) * Normal: Outward normal points in the $-y$ direction, so $\hat{\mathbf{n}} = -\hat{\mathbf{j}}$. * **Area Element:** $dS = dx dz$. * **Integrand:** $\mathbf{F} \cdot \hat{\mathbf{n}} = y^2$. Since $y = 0$, $\mathbf{F} \cdot \hat{\mathbf{n}} = 0$. * **Integral:**

$$I_4 = 0$$

5. Top Face ($z = a$) * Normal: Outward normal points in the $+z$ direction, so $\hat{\mathbf{n}} = \hat{\mathbf{k}}$. * **Area Element:** $dS = dx dy$. * **Integrand:** $\mathbf{F} \cdot \hat{\mathbf{n}} = yz$. Since $z = a$, $\mathbf{F} \cdot \hat{\mathbf{n}} = ay$. * **Integral:**

$$I_5 = \int_0^a \int_0^a ay dx dy = a \int_0^a y dy \int_0^a dx = a \left(\frac{a^2}{2} \right) (a) = \frac{a^4}{2}$$

6. Bottom Face ($z = 0$) * Normal: Outward normal points in the $-z$ direction, so $\hat{\mathbf{n}} = -\hat{\mathbf{k}}$. * **Area Element:** $dS = dx dy$. * **Integrand:** $\mathbf{F} \cdot \hat{\mathbf{n}} = -yz$. Since $z = 0$, $\mathbf{F} \cdot \hat{\mathbf{n}} = 0$. * **Integral:**

$$I_6 = 0$$

Total Integral: Summing the results from all 6 faces:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS = 2a^4 + 0 - a^4 + 0 + \frac{a^4}{2} + 0 = a^4 + \frac{a^4}{2} = \frac{3a^4}{2}$$

9.2.2 Problem 2: Volume Integral Over a Solid Cylinder

Question: Evaluate $\iiint_V (\nabla \cdot \mathbf{F}) dV$ where $\mathbf{F} = 4x\hat{\mathbf{i}} - 2y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}$, and V is the solid region bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = 0$ and $z = 3$.

Solution: Step 1: Compute the Integrand ($\nabla \cdot \mathbf{F}$) Calculate the divergence of the vector field:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

Step 2: Determine Limits of Integration The volume V is a solid cylinder. The height is bounded by the planes $z = 0$ and $z = 3$. The base of the cylinder on the xy -plane is the circle $x^2 + y^2 = 4$ (radius $r = 2$).

- **Limits for z :** Direct boundary conditions give $z = 0$ to $z = 3$. - **Limits for x and y :** From the base circle $x^2 + y^2 = 4$, x varies from -2 to 2 . For a given x , y varies from the lower half of the circle to the upper half, so y goes from $-\sqrt{4-x^2}$ to $\sqrt{4-x^2}$.

Step 3: Setup and Evaluate the Triple Integral

$$I = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^3 (4 - 4y + 2z) dz dy dx$$

Inner Integration (with respect to z):

$$\int_0^3 (4 - 4y + 2z) dz = [4z - 4yz + z^2]_0^3 = (12 - 12y + 9) - 0 = 21 - 12y = 3(7 - 4y)$$

Middle Integration (with respect to y): Substitute the result back:

$$\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 3(7 - 4y) dy = 3 [7y - 2y^2]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$$

Apply the upper and lower limits:

$$= 3 \left[\left(7\sqrt{4-x^2} - 2(4-x^2) \right) - \left(7(-\sqrt{4-x^2}) - 2(4-x^2) \right) \right]$$

The squared terms cancel out:

$$= 3 \left[14\sqrt{4-x^2} \right] = 42\sqrt{4-x^2}$$

Outer Integration (with respect to x): Substitute the result back:

$$I = \int_{-2}^2 42\sqrt{4-x^2} dx$$

Use the standard integral formula: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$. Here $a = 2$, so $a^2 = 4$.

$$I = 42 \left[\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) \right]_{-2}^2$$

When evaluating at $x = 2$ and $x = -2$, the $\sqrt{4-x^2}$ term becomes 0.

$$= 42 \left[(0 + 2 \sin^{-1}(1)) - (0 + 2 \sin^{-1}(-1)) \right]$$

Since $\sin^{-1}(1) = \frac{\pi}{2}$ and $\sin^{-1}(-1) = -\frac{\pi}{2}$:

$$= 42 \left[2 \left(\frac{\pi}{2} \right) - 2 \left(-\frac{\pi}{2} \right) \right] = 42[\pi - (-\pi)] = 42(2\pi) = 84\pi$$

Final Answer: 84π

9.3 Key Takeaways

- **Closed Surfaces:** When evaluating a surface integral over a closed shape like a cube, you must evaluate the integral over each individual face and sum the results. The normal vector $\hat{\mathbf{n}}$ must *always* point outward from the enclosed volume.
- **Volume Integration:** To perform a volume integral over a symmetric solid like a cylinder, start by identifying the constant height bounds for z . Then, project the solid onto the xy -plane to determine the limits for x and y , which often involves circle boundaries.
- **Sequential Integration:** Always integrate variables with functional limits (like $y = f(x)$) before variables with constant limits.

9.4 What Comes Next

With a firm understanding of Line, Surface, and Volume Integrals, we are now fully equipped to tackle the monumental theorems of Vector Calculus: **Gauss' Divergence Theorem**, **Stokes' Theorem**, and **Green's Theorem**.

10 Lecture 10: Concept of Stokes' Theorem (Part 1)

Series: Vector Calculus Lecture: 10 of 14 Video: <https://www.youtube.com/watch?v=uuWw4hG0s3Q>

10.1 Prerequisites

We have extensively studied both Line Integrals $\oint_C \mathbf{F} \cdot d\mathbf{r}$ and Surface Integrals $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$. We are now ready to establish a powerful relationship between these two types of integrals using **Stokes' Theorem**.

10.2 Stokes' Theorem

10.2.1 Statement

Let S be an open, two-sided surface bounded by a closed, non-intersecting curve C . If a vector point function $\mathbf{F}$ is continuous and differentiable over a region containing S and C , then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

Where: $\oint_C$ denotes the line integral around the closed boundary curve C . $\iint_S$ denotes the surface integral over the surface S bounded by C . $\nabla \times \mathbf{F}$ is the curl of the vector field $\mathbf{F}$. $\hat{\mathbf{n}}$ is the outward unit normal vector to the surface S .

10.2.2 Orientation (The Right-Hand Screw Rule)

The theorem requires a strict convention between the direction you traverse the boundary curve C and the direction of the outward unit normal $\hat{\mathbf{n}}$.

If you traverse the boundary curve C in the **anticlockwise (positive) direction**, and you curl the fingers of your right hand in that direction, your thumb points in the direction of the outward unit normal $\hat{\mathbf{n}}$. For example, if a surface lies flat in the xy -plane and you traverse its boundary anticlockwise, the outward normal points "upward" along the positive z -axis ($\hat{\mathbf{n}} = \hat{\mathbf{k}}$).

10.2.3 Why is this Theorem Useful?

Stokes' Theorem allows you to swap a difficult line integral for an easier surface integral, or vice versa. For example, integrating a vector field along a rectangular boundary requires calculating 4 separate line integrals and summing them up. Using Stokes' theorem, you can simply calculate the curl of the vector field and evaluate a *single* surface integral over the flat rectangular area.

10.3 Solved Problems

10.3.1 Problem 1: Verifying Stokes' Theorem on a Rectangle

Question: Verify Stokes' Theorem for $\mathbf{F} = (x^2 + y^2)\hat{\mathbf{i}} - 2xy\hat{\mathbf{j}}$ taken round the rectangle bounded by $x = \pm a, y = 0, y = b$.

Solution: To "verify" the theorem, we must independently calculate both the Left-Hand Side (Line Integral) and the Right-Hand Side (Surface Integral) and prove they yield the exact same result.

10.3.1.1 Part 1: Evaluating the Line Integral (LHS) Let's define the rectangular boundary C in the xy -plane with vertices: $C(-a, 0)$, $B(a, 0)$, $A(a, b)$, and $D(-a, b)$. We will traverse this boundary in the **anticlockwise direction**. This divides the boundary into 4 line segments: 1. C_3 (**Bottom**): From $C(-a, 0)$ to $B(a, 0)$. Here, $y = 0 \implies dy = 0$. x goes from $-a$ to a . 2. C_4 (**Right**): From $B(a, 0)$ to $A(a, b)$. Here, $x = a \implies dx = 0$. y goes from 0 to b . 3. C_1 (**Top**): From $A(a, b)$ to $D(-a, b)$. Here, $y = b \implies dy = 0$. x goes from a to $-a$. 4. C_2 (**Left**): From $D(-a, b)$ to $C(-a, 0)$. Here, $x = -a \implies dx = 0$. y goes from b to 0 .

The general line integrand is $\mathbf{F} \cdot d\mathbf{r} = (x^2 + y^2)dx - 2xy dy$. Let's evaluate this for each segment:

On C_3 (Bottom):

$$I_3 = \int_{-a}^a (x^2 + 0) dx = \left[\frac{x^3}{3} \right]_{-a}^a = \frac{a^3}{3} - \left(-\frac{a^3}{3} \right) = \frac{2a^3}{3}$$

On C_4 (Right):

$$I_4 = \int_0^b -2(a)y dy = -2a \left[\frac{y^2}{2} \right]_0^b = -ab^2$$

On C_1 (Top):

$$I_1 = \int_a^{-a} (x^2 + b^2) dx = \left[\frac{x^3}{3} + b^2 x \right]_a^{-a} = \left(-\frac{a^3}{3} - ab^2 \right) - \left(\frac{a^3}{3} + ab^2 \right) = -\frac{2a^3}{3} - 2ab^2$$

On C_2 (Left):

$$I_2 = \int_b^0 -2(-a)y dy = 2a \int_b^0 y dy = 2a \left[\frac{y^2}{2} \right]_b^0 = 2a \left(0 - \frac{b^2}{2} \right) = -ab^2$$

Summing the four line integrals gives the total LHS:

$$\text{LHS} = I_1 + I_2 + I_3 + I_4 = \left(-\frac{2a^3}{3} - 2ab^2 \right) - ab^2 + \left(\frac{2a^3}{3} \right) - ab^2 = -4ab^2$$

10.3.1.2 Part 2: Evaluating the Surface Integral (RHS) We need to compute $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$ over the rectangular area S .

First, compute the curl $(\nabla \times \mathbf{F})$:

$$\begin{aligned} \nabla \times \mathbf{F} &= \det \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix} \\ &= \hat{\mathbf{i}}(0 - 0) - \hat{\mathbf{j}}(0 - 0) + \hat{\mathbf{k}} \left(\frac{\partial}{\partial x}(-2xy) - \frac{\partial}{\partial y}(x^2 + y^2) \right) \\ &= \hat{\mathbf{k}}(-2y - 2y) = -4y\hat{\mathbf{k}} \end{aligned}$$

Because the surface lies flat in the xy -plane and we traversed the boundary anticlockwise, the outward unit normal points along the positive z -axis: $\hat{\mathbf{n}} = \hat{\mathbf{k}}$.

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = (-4y\hat{\mathbf{k}}) \cdot \hat{\mathbf{k}} = -4y$$

Now, set up the double integral over the rectangular bounds (x from $-a$ to a , and y from 0 to b):

$$\text{RHS} = \int_{-a}^a \int_0^b -4y dy dx$$

Integrate with respect to y :

$$\int_0^b -4y dy = -4 \left[\frac{y^2}{2} \right]_0^b = -2b^2$$

Integrate the result with respect to x :

$$\int_{-a}^a -2b^2 dx = -2b^2 [x]_{-a}^a = -2b^2(a - (-a)) = -2b^2(2a) = -4ab^2$$

10.3.1.3 Conclusion We have independently evaluated both sides:

$$\text{LHS} = -4ab^2 \quad \text{and} \quad \text{RHS} = -4ab^2$$

Since $\text{LHS} = \text{RHS}$, Stokes' Theorem is strictly verified. Notice how much faster the surface integral approach (RHS) was compared to computing four separate line integrals (LHS).

10.4 What Comes Next

11 Lecture 11: Concept of Stokes' Theorem (Part 2)

Series: Vector Calculus **Lecture:** 11 of 14 **Video:** <https://www.youtube.com/watch?v=P74t2Ix5NZI>

11.1 Prerequisites

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$$

For a rectangular boundary on a 2D plane. In this lecture, we will verify the theorem for a circular boundary and, most importantly, for a curved 3D surface (a hemisphere) to show that the theorem holds regardless of the surface geometry.

11.2 Solved Problems

11.2.1 Problem 1: Verifying Stokes' Theorem on a Unit Circle

Question: Verify Stokes' theorem for the function $\mathbf{F} = (x + 2y)\hat{\mathbf{i}} + (y + 3x)\hat{\mathbf{j}}$ where C is the unit circle in the xy -plane.

Solution: A unit circle in the xy -plane has the equation $x^2 + y^2 = 1$ and lies at $z = 0$.

11.2.1.1 Part 1: Evaluating the Line Integral (LHS) Since $z = 0$, $dz = 0$. The line integral becomes:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C [(x + 2y)dx + (y + 3x)dy]$$

To evaluate this around the circular boundary, we convert to parametric coordinates. Let ϕ be the angle parameter from 0 to 2π : * $x = \cos \phi \implies dx = -\sin \phi d\phi$ * $y = \sin \phi \implies dy = \cos \phi d\phi$

Substitute these into the integral:

$$\begin{aligned} & \int_0^{2\pi} [(\cos \phi + 2 \sin \phi)(-\sin \phi) + (\sin \phi + 3 \cos \phi)(\cos \phi)] d\phi \\ &= \int_0^{2\pi} (-\sin \phi \cos \phi - 2 \sin^2 \phi + \sin \phi \cos \phi + 3 \cos^2 \phi) d\phi \end{aligned}$$

The $-\sin \phi \cos \phi$ terms cancel:

$$= \int_0^{2\pi} (3 \cos^2 \phi - 2 \sin^2 \phi) d\phi$$

Using the identity $\sin^2 \phi = 1 - \cos^2 \phi$:

$$= \int_0^{2\pi} (3 \cos^2 \phi - 2(1 - \cos^2 \phi)) d\phi = \int_0^{2\pi} (5 \cos^2 \phi - 2) d\phi$$

Convert using the double angle formula $\cos^2 \phi = \frac{1 + \cos 2\phi}{2}$:

$$= \int_0^{2\pi} \left(\frac{5}{2}(1 + \cos 2\phi) - 2 \right) d\phi = \int_0^{2\pi} \left(\frac{1}{2} + \frac{5}{2} \cos 2\phi \right) d\phi$$

Integrate:

$$\left[\frac{\phi}{2} + \frac{5}{4} \sin 2\phi \right]_0^{2\pi} = \left(\frac{2\pi}{2} + 0 \right) - (0 + 0) = \pi$$

So, LHS = π .

11.2.1.2 Part 2: Evaluating the Surface Integral (RHS) We evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$ over the flat unit disk bounded by the circle. First, compute the curl:

$$\nabla \times \mathbf{F} = \det \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y & y+3x & 0 \end{vmatrix} = \hat{\mathbf{k}} \left(\frac{\partial}{\partial x}(y+3x) - \frac{\partial}{\partial y}(x+2y) \right) = \hat{\mathbf{k}}(3-2) = \hat{\mathbf{k}}$$

Since the surface is flat in the xy -plane, the outward unit normal is $\hat{\mathbf{n}} = \hat{\mathbf{k}}$.

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$$

The integral becomes:

$$\iint_S 1 dS = \text{Area}(S)$$

Since S is a unit circle ($r = 1$), its area is $\pi r^2 = \pi$.

So, $\text{RHS} = \pi$. **LHS = RHS**, hence verified.

11.2.2 Problem 2: Verifying Stokes' Theorem on a Hemisphere

Question: Verify Stokes' theorem for $\mathbf{F} = y\hat{\mathbf{i}} + z\hat{\mathbf{j}} + x\hat{\mathbf{k}}$, where S is the upper hemisphere $x^2 + y^2 + z^2 = 1$ for $z \geq 0$, and C is its boundary.

Solution: The surface S is a 3D curved hemisphere dome. The boundary C is the base ring of this dome lying in the xy -plane, which is exactly the unit circle $x^2 + y^2 = 1$ at $z = 0$.

11.2.2.1 Part 1: Evaluating the Line Integral (LHS) Since the boundary C lies entirely in the xy -plane, $z = 0$ and $dz = 0$. The integral $\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (y dx + z dy + x dz)$ simplifies to:

$$\oint_C y dx$$

Using parametric equations for the unit circle $x = \cos \phi \implies dx = -\sin \phi d\phi$, and $y = \sin \phi$:

$$\begin{aligned} \int_0^{2\pi} \sin \phi (-\sin \phi) d\phi &= - \int_0^{2\pi} \sin^2 \phi d\phi = - \int_0^{2\pi} \frac{1 - \cos 2\phi}{2} d\phi \\ &= -\frac{1}{2} \left[\phi - \frac{\sin 2\phi}{2} \right]_0^{2\pi} = -\frac{1}{2} (2\pi - 0) = -\pi \end{aligned}$$

So, $\text{LHS} = -\pi$.

11.2.2.2 Part 2: Evaluating the Surface Integral (RHS) We evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$ over the curved hemispherical surface. Compute the curl:

$$\nabla \times \mathbf{F} = \det \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = \hat{\mathbf{i}}(0-1) - \hat{\mathbf{j}}(1-0) + \hat{\mathbf{k}}(0-1) = -\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}$$

Find the unit normal vector $\hat{\mathbf{n}}$ for the sphere $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$:

$$\nabla g = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}} \implies \hat{\mathbf{n}} = \frac{\nabla g}{|\nabla g|} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

Compute the dot product:

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = (-\hat{\mathbf{i}} - \hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) = -(x + y + z)$$

We convert this to spherical polar coordinates to integrate over the curved surface ($r = 1$): $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$, $z = \cos \theta$, $dS = \sin \theta d\theta d\phi$. Limits: ϕ from 0 to 2π (full circle), θ from 0 to $\pi/2$ (upper hemisphere only).

Set up the integral:

$$\text{RHS} = \int_0^{2\pi} \int_0^{\pi/2} -(\sin \theta \cos \phi + \sin \theta \sin \phi + \cos \theta) \sin \theta d\theta d\phi$$

Integrate with respect to ϕ first. Over $[0, 2\pi]$, the integral of $\cos \phi$ is 0, and the integral of $\sin \phi$ is 0. The only term that survives is the $\cos \theta$ term:

$$\int_0^{2\pi} \cos \theta \sin \theta d\phi = 2\pi \cos \theta \sin \theta$$

Now integrate with respect to θ :

$$\begin{aligned} - \int_0^{\pi/2} 2\pi \cos \theta \sin \theta d\theta &= -\pi \int_0^{\pi/2} 2 \sin \theta \cos \theta d\theta = -\pi \int_0^{\pi/2} \sin 2\theta d\theta \\ &= -\pi \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2} = \frac{\pi}{2} (\cos \pi - \cos 0) = \frac{\pi}{2} (-1 - 1) = -\pi \end{aligned}$$

So, $\text{RHS} = -\pi$. **LHS = RHS**, hence verified.

11.2.3 Practice Application

Question: Evaluate $\oint_C (xy dx + xy^2 dy)$ taken round the positively oriented square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$ by using Stokes' theorem.

Solution Approach: Instead of evaluating 4 separate line integrals for the square boundary, we use Stokes' Theorem to evaluate the surface integral: 1. $\mathbf{F} = xy\hat{\mathbf{i}} + xy^2\hat{\mathbf{j}}$. 2. $\nabla \times \mathbf{F} = (y^2 - x)\hat{\mathbf{k}}$. 3. $\hat{\mathbf{n}} = \hat{\mathbf{k}}$ (square lies in xy -plane). 4. Integral: $\iint_S (y^2 - x) dx dy$. By symmetry, the $\iint_S x dx dy$ component over the centered square is 0. The remaining integral $4 \int_0^1 \int_0^{1-x} y^2 dy dx$ evaluates to exactly $\frac{1}{3}$.

11.3 What Comes Next

We have completed our study of Stokes' Theorem. In the next phase, we will move on to the **Gauss Divergence Theorem**, which relates surface integrals to volume integrals.

12 Lecture 12: Concept of Gauss Divergence Theorem (Part 1)

Series: Vector Calculus Lecture: 12 of 14 Video: <https://www.youtube.com/watch?v=vZas0kR9cUY>

12.1 Prerequisites

We have already seen how **Stokes' Theorem** relates line integration to surface integration. Now, we introduce the **Gauss Divergence Theorem**, which provides a profound connection between **surface integrals** and **volume integrals**. A strong foundation in double and triple integration bounds is essential here.

12.2 Gauss Divergence Theorem

12.2.1 Statement

Let V be a solid volume bounded by a closed surface S . If $\mathbf{F}$ is a continuously differentiable vector point function, then the surface integral of $\mathbf{F}$ over S is equal to the volume integral of the divergence of $\mathbf{F}$ over V :

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

Where: $\iint_S$ denotes the surface integral over the *closed* surface S . $\iiint_V$ denotes the triple (volume) integral over the enclosed space V . $\hat{\mathbf{n}}$ is the outward-pointing unit normal vector to the surface S . $\nabla \cdot \mathbf{F}$ is the divergence (a scalar) of the vector field $\mathbf{F}$.

12.2.2 Why is this Theorem Useful?

Evaluating a surface integral over a fully closed shape (like a cube, sphere, or tetrahedron) requires calculating the flux across every single bounding face separately. The Gauss Divergence Theorem allows us to bypass this tedious calculation entirely. Instead of integrating over multiple external surfaces, we compute the divergence of the vector field and evaluate a single triple integral over the solid interior volume.

12.3 Solved Problems

12.3.1 Problem 1: Integration over a Tetrahedron

Question: Evaluate the surface integral $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS$, where $\mathbf{F} = (x + y^2)\hat{\mathbf{i}} - 2x\hat{\mathbf{j}} + 2yz\hat{\mathbf{k}}$, and S is the closed surface bounded by the coordinate planes ($x = 0, y = 0, z = 0$) and the plane $2x + y + 2z = 6$.

Solution: The surface S is a tetrahedron. Evaluating this directly would require 4 separate double integrals (one for each of the 4 triangular faces). Instead, we will use the Gauss Divergence Theorem to evaluate a single volume integral over the interior space V .

Step 1: Compute the Divergence

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x + y^2) + \frac{\partial}{\partial y}(-2x) + \frac{\partial}{\partial z}(2yz) = 1 + 0 + 2y = 1 + 2y$$

The volume integral is: $\iiint_V (1 + 2y) dV$.

Step 2: Determine Limits of Integration We find the intercepts of the plane $2x + y + 2z = 6$: x -intercept (set $y = z = 0$): $2x = 6 \implies x = 3$ y -intercept (set $x = z = 0$): $y = 6$ z -intercept (set $x = y = 0$): $2z = 6 \implies z = 3$

The limits are established by working from the top down to the base projection: 1. **z -limits:** From the floor ($z = 0$) to the slanted roof ($z = \frac{6-2x-y}{2}$). 2. **y -limits:** On the floor projection ($z = 0$), the plane becomes the line $2x + y = 6 \implies y = 6 - 2x$. So, y goes from 0 to $6 - 2x$. 3. **x -limits:** From the origin to the x -intercept ($x = 3$). So, x goes from 0 to 3.

Step 3: Evaluate the Triple Integral

$$I = \int_0^3 \int_0^{6-2x} \int_0^{\frac{6-2x-y}{2}} (1+2y) \, dz \, dy \, dx$$

Inner Integral (w.r.t z):

$$\int_0^{\frac{6-2x-y}{2}} (1+2y) \, dz = (1+2y)[z]_0^{\frac{6-2x-y}{2}} = (1+2y) \left(\frac{6-2x-y}{2} \right)$$

Expanding this expression gives:

$$\frac{1}{2}(6-2x-y+12y-4xy-2y^2) = \frac{1}{2}(6-2x+11y-4xy-2y^2)$$

Middle Integral (w.r.t y):

$$\frac{1}{2} \int_0^{6-2x} (6-2x+11y-4xy-2y^2) \, dy = \frac{1}{2} \left[(6-2x)y + \frac{11}{2}y^2 - 2xy^2 - \frac{2}{3}y^3 \right]_0^{6-2x}$$

Substitute the upper limit $y = 6 - 2x$:

$$= \frac{1}{2} \left[(6-2x)^2 + \frac{11}{2}(6-2x)^2 - 2x(6-2x)^2 - \frac{2}{3}(6-2x)^3 \right]$$

Factor out $(6-2x)^2$:

$$= \frac{1}{2}(6-2x)^2 \left(1 + \frac{11}{2} - 2x - \frac{2}{3}(6-2x) \right) = \dots (\text{after algebraic expansion}) = -\frac{4}{3}x^3 + 13x^2 - 42x + 45$$

Outer Integral (w.r.t x):

$$\begin{aligned} \int_0^3 \left(-\frac{4}{3}x^3 + 13x^2 - 42x + 45 \right) dx &= \left[-\frac{1}{3}x^4 + \frac{13}{3}x^3 - 21x^2 + 45x \right]_0^3 \\ &= -\frac{1}{3}(81) + \frac{13}{3}(27) - 21(9) + 45(3) = -27 + 117 - 189 + 135 = 36 \end{aligned}$$

[!NOTE] **Correction:** The original lecture contained an algebraic expansion error here. It incorrectly simplified the polynomial to $-4x^3$ (missing the division by 3) and subsequently forced a flawed arithmetic addition $(-81 + 117 - 189 + 135)$ to arrive at the desired final answer of 36. The mathematical steps have been corrected above to ensure strict logical integrity.

Final Answer: 36**12.3.2 Problem 2: Integration over a Sphere**

Question: Evaluate the surface integral $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$, where $\mathbf{F} = x\hat{\mathbf{i}} - y\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}$, and S is the boundary of the sphere $x^2 + y^2 + (z-1)^2 = 1$.

Solution: We use Gauss Divergence Theorem to convert this to a volume integral.

Step 1: Compute Divergence

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(2z) = 1 - 1 + 2 = 2$$

Step 2: Evaluate the Volume Integral The integral over the spherical volume V becomes:

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \iiint_V 2 \, dV = 2 \iiint_V dV$$

Notice that $\iiint_V dV$ is simply the geometric volume of the solid region V . The region is a sphere with equation $x^2 + y^2 + (z - 1)^2 = 1$, which has a radius $R = 1$ and is centered at $(0, 0, 1)$.

The geometric volume of a sphere is $V = \frac{4}{3}\pi R^3$. For $R = 1$:

$$V = \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$$

Substitute this back into the expression:

$$2 \iiint_V dV = 2(V) = 2\left(\frac{4}{3}\pi\right) = \frac{8}{3}\pi$$

Final Answer: $\frac{8}{3}\pi$

[!NOTE] This problem demonstrates a massive shortcut: if the divergence $\nabla \cdot \mathbf{F}$ is a constant scalar, you do not need to set up complex integration bounds. You can simply multiply that constant by the known geometric volume formula of the enclosed solid!

12.4 What Comes Next

13 Lecture 13: Concept of Gauss Divergence Theorem (Part 2)

Series: Vector Calculus **Lecture:** 13 of 14 **Video:** <https://www.youtube.com/watch?v=IjWYwfuPBKU>

13.1 Prerequisites

We previously used the Gauss Divergence Theorem:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

To evaluate surface integrals by converting them into simpler volume integrals. In this lecture, we will perform a full **verification** of the theorem over a composite surface (a cylinder) and explore a neat trick utilizing the center of gravity to solve cone integrals instantly.

13.2 Solved Problems

13.2.1 Problem 1: Verifying Gauss Divergence Theorem on a Cylinder

Question: Verify the divergence theorem for $\mathbf{F} = 4x\hat{\mathbf{i}} - 2y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}$ taken over the region bounded by $x^2 + y^2 = 4$, $z = 0$, and $z = 3$.

Solution: The region is a solid cylinder of radius $R = 2$ and height $h = 3$. To verify the theorem, we must evaluate both the volume integral (RHS) and the surface integral (LHS) and ensure they are equal.

13.2.1.1 Part 1: Evaluating the Volume Integral (RHS) First, calculate the divergence of $\mathbf{F}$:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) = 4 - 4y + 2z$$

The triple integral is $\iiint_V (4 - 4y + 2z) \, dx \, dy \, dz$.

The bounds for the cylinder are: * z from 0 to 3. * y from $-\sqrt{4-x^2}$ to $\sqrt{4-x^2}$ (from the circular base). * x from -2 to 2 .

Integrate with respect to z :

$$\int_0^3 (4 - 4y + 2z) \, dz = [4z - 4yz + z^2]_0^3 = 12 - 12y + 9 = 21 - 12y = 3(7 - 4y)$$

Integrate with respect to y :

$$3 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (7 - 4y) \, dy = 3 [7y - 2y^2]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}$$

Because y^2 is an even function, the subtraction of limits causes the y^2 terms to cancel out entirely.

$$= 3 \left(14\sqrt{4-x^2} \right) = 42\sqrt{4-x^2}$$

Integrate with respect to x :

$$I_V = \int_{-2}^2 42\sqrt{4-x^2} \, dx$$

Using the standard integration formula $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$:

$$I_V = 42 \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) \right]_{-2}^2$$

When evaluated at $x = \pm 2$, the square root term becomes 0.

$$= 42 \left[2 \sin^{-1}(1) - 2 \sin^{-1}(-1) \right] = 42 \left(2 \left(\frac{\pi}{2} \right) - 2 \left(-\frac{\pi}{2} \right) \right) = 42(\pi + \pi) = 84\pi$$

So, RHS = 84π .

13.2.1.2 Part 2: Evaluating the Surface Integral (LHS) The closed surface S consists of 3 parts: the bottom base (S_1), the top base (S_2), and the curved lateral surface (S_3).

1. Bottom Base (S_1 at $z = 0$): * Outward normal: $\hat{\mathbf{n}}_1 = -\hat{\mathbf{k}}$ * $\mathbf{F} \cdot \hat{\mathbf{n}}_1 = -z^2$. * Since $z = 0$ on this surface, the integral is exactly 0.

2. Top Base (S_2 at $z = 3$): * Outward normal: $\hat{\mathbf{n}}_2 = \hat{\mathbf{k}}$ * $\mathbf{F} \cdot \hat{\mathbf{n}}_2 = z^2$. * Since $z = 3$, the integrand is $3^2 = 9$. * The integral is $9 \times \text{Area}(S_2) = 9 \times (\pi \cdot 2^2) = 36\pi$.

3. Curved Lateral Surface (S_3 where $x^2 + y^2 = 4$): * Gradient of the surface $g = x^2 + y^2 - 4 = 0$ is $\nabla g = 2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}$. * Outward normal $\hat{\mathbf{n}}_3 = \frac{2x\hat{\mathbf{i}} + 2y\hat{\mathbf{j}}}{\sqrt{4x^2 + 4y^2}} = \frac{x\hat{\mathbf{i}} + y\hat{\mathbf{j}}}{2}$. * $\mathbf{F} \cdot \hat{\mathbf{n}}_3 = (4x\hat{\mathbf{i}} - 2y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}) \cdot \frac{x\hat{\mathbf{i}} + y\hat{\mathbf{j}}}{2} = 2x^2 - y^3$.
* Convert to cylindrical coordinates: $x = 2 \cos \theta$, $y = 2 \sin \theta$, $dS_3 = R d\theta dz = 2 d\theta dz$. * Limits: $\theta \in [0, 2\pi]$ and $z \in [0, 3]$.

$$\begin{aligned} \int_0^{2\pi} \int_0^3 (2(2 \cos \theta)^2 - (2 \sin \theta)^3) 2 dz d\theta &= \int_0^{2\pi} \int_0^3 (8 \cos^2 \theta - 8 \sin^3 \theta) 2 dz d\theta \\ &= 48 \int_0^{2\pi} (\cos^2 \theta - \sin^3 \theta) d\theta \end{aligned}$$

Since $\int_0^{2\pi} \sin^3 \theta d\theta = 0$ over a full period, and $\int_0^{2\pi} \cos^2 \theta d\theta = \pi$:

$$= 48(\pi - 0) = 48\pi$$

Total Surface Integral:

$$I_S = I_{S_1} + I_{S_2} + I_{S_3} = 0 + 36\pi + 48\pi = 84\pi$$

So, LHS = 84π .

Since LHS = RHS = 84π , the Gauss Divergence Theorem is fully verified!

13.2.2 Problem 2: Center of Gravity Trick for Cones

Question: Evaluate $\iint_S (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}) \cdot \hat{\mathbf{n}} dS$, where S is the closed surface bounded by the cone $x^2 + y^2 = z^2$ and the plane $z = 1$.

Solution: Using the Divergence Theorem, we convert this to a volume integral.

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z^2) = 1 + 1 + 2z = 2(1 + z)$$

The volume integral is:

$$\iiint_V 2(1 + z) dV = 2 \iiint_V dV + 2 \iiint_V z dV$$

We recognize two geometric properties: 1. $\iiint_V dV$ is the geometric volume V of the cone. 2. By definition, the z -coordinate of the centroid (center of gravity) is $\bar{z} = \frac{\iiint_V z dV}{V}$. Therefore, $\iiint_V z dV = V\bar{z}$.

Substitute these in:

$$2V + 2V\bar{z} = 2V(1 + \bar{z})$$

Now, we simply use the known geometric formulas for a solid cone: * The cone has a height $h = 1$ and a base radius $R = 1$ (since $x^2 + y^2 = z^2$ at $z = 1$). * The volume of a cone is $V = \frac{1}{3}\pi R^2 h = \frac{1}{3}\pi(1)^2(1) = \frac{1}{3}\pi$. * The center of gravity of a solid right circular cone lies on its axis at $\bar{z} = \frac{3}{4}h$. Since $h = 1$, $\bar{z} = \frac{3}{4}$.

Substitute these known values:

$$\text{Result} = 2 \left(\frac{1}{3}\pi \right) \left(1 + \frac{3}{4} \right) = \frac{2}{3}\pi \left(\frac{7}{4} \right) = \frac{7}{6}\pi$$

[!TIP] If a divergence evaluates to a linear coordinate (like z), you can often solve the integral instantly by utilizing the formula $V\bar{z}$, bypassing complex integration setups completely!

13.3 What Comes Next

We have completed our study of the Gauss Divergence Theorem. In the final lecture, we will cover **Green's Theorem**, which relates line integrals to double area integrals in the 2D plane.

14 Lecture 14: Concept of Green's Theorem

Series: Vector Calculus **Lecture:** 14 of 14 **Video:** <https://www.youtube.com/watch?v=EiWb17ofMnA>

14.1 Prerequisites

We have completed our study of Stokes' Theorem (relating 3D line integrals to 3D surface integrals) and Gauss Divergence Theorem (relating 3D surface integrals to 3D volume integrals). We conclude our vector calculus series with **Green's Theorem**, which is essentially a specialized 2D case of Stokes' Theorem. It relates a line integral around a simple closed curve in a 2D plane to a double integral over the flat area enclosed by that curve.

14.2 Green's Theorem in a Plane

14.2.1 Statement

If $P(x, y)$ and $Q(x, y)$ are continuous functions with continuous partial derivatives in a region S enclosed by a closed curve C in the xy -plane, then:

$$\oint_C (P dx + Q dy) = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Where: * C is traversed in the positive (anticlockwise) direction. * P is the expression multiplied by dx . * Q is the expression multiplied by dy .

14.2.2 Why is this Theorem Useful?

Just like the other theorems, Green's Theorem allows us to bypass tedious boundary integrations. If a boundary C consists of multiple line segments (like a triangle or a rectangle), evaluating $\oint_C$ directly requires calculating multiple separate integrals. By applying Green's Theorem, you can simply evaluate a single double integral over the bounded flat region S .

14.3 Solved Problems

14.3.1 Problem 1: Evaluating a Line Integral via Double Integral

Question: Evaluate by Green's Theorem the line integral $\oint_C (e^{-x} \sin y dx + e^{-x} \cos y dy)$, where C is the rectangle with vertices $(0, 0), (\pi, 0), (\pi, \frac{\pi}{2}), (0, \frac{\pi}{2})$.

Solution: We identify the components P and Q from the integral form $\oint_C (P dx + Q dy)$: * $P = e^{-x} \sin y \implies \frac{\partial P}{\partial y} = e^{-x} \cos y$ * $Q = e^{-x} \cos y \implies \frac{\partial Q}{\partial x} = -e^{-x} \cos y$

Compute the integrand for the double integral:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -e^{-x} \cos y - e^{-x} \cos y = -2e^{-x} \cos y$$

Set up the double integral over the rectangular region S . The limits are clearly defined by the vertices: x varies from 0 to π , and y varies from 0 to $\frac{\pi}{2}$.

$$\iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_0^\pi \int_0^{\frac{\pi}{2}} -2e^{-x} \cos y dy dx$$

Since the limits are constants, we can separate the variables:

$$= -2 \left(\int_0^\pi e^{-x} dx \right) \left(\int_0^{\frac{\pi}{2}} \cos y dy \right)$$

Evaluate the y -integral:

$$\int_0^{\frac{\pi}{2}} \cos y \, dy = [\sin y]_0^{\frac{\pi}{2}} = 1 - 0 = 1$$

Evaluate the x -integral:

$$\int_0^{\pi} e^{-x} \, dx = [-e^{-x}]_0^{\pi} = -e^{-\pi} - (-1) = 1 - e^{-\pi}$$

Multiply them together with the constant -2 :

$$= -2(1 - e^{-\pi})(1) = 2(e^{-\pi} - 1)$$

Final Answer: $2(e^{-\pi} - 1)$

14.3.2 Problem 2: Verifying Green's Theorem on a Circle

Question: Verify Green's Theorem for the line integral $\oint_C (x + 2y) \, dx + (y + 3x) \, dy$ where C is the unit circle $x^2 + y^2 = 1$.

Solution: To verify, we must independently compute both the LHS (Line Integral) and the RHS (Double Integral).

Part 1: Line Integral (LHS) Convert the circle $x^2 + y^2 = 1$ to parametric coordinates: $x = \cos \theta$, $dx = -\sin \theta \, d\theta$ and $y = \sin \theta$, $dy = \cos \theta \, d\theta$, for $\theta \in [0, 2\pi]$.

$$\begin{aligned} \oint_C &= \int_0^{2\pi} [(\cos \theta + 2 \sin \theta)(-\sin \theta) + (\sin \theta + 3 \cos \theta)(\cos \theta)] \, d\theta \\ &= \int_0^{2\pi} (-\sin \theta \cos \theta - 2 \sin^2 \theta + \sin \theta \cos \theta + 3 \cos^2 \theta) \, d\theta \end{aligned}$$

The $\sin \theta \cos \theta$ terms cancel:

$$= \int_0^{2\pi} (3 \cos^2 \theta - 2 \sin^2 \theta) \, d\theta$$

Using double-angle formulas ($\cos^2 \theta = \frac{1+\cos 2\theta}{2}$ and $\sin^2 \theta = \frac{1-\cos 2\theta}{2}$):

$$\begin{aligned} &= \int_0^{2\pi} \left[3 \left(\frac{1+\cos 2\theta}{2} \right) - 2 \left(\frac{1-\cos 2\theta}{2} \right) \right] \, d\theta = \int_0^{2\pi} \left(\frac{1}{2} + \frac{5}{2} \cos 2\theta \right) \, d\theta \\ &= \left[\frac{\theta}{2} + \frac{5}{4} \sin 2\theta \right]_0^{2\pi} = \frac{2\pi}{2} + 0 = \pi \end{aligned}$$

LHS = π

Part 2: Double Integral (RHS) Identify $P = x + 2y$ and $Q = y + 3x$.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3 - 2 = 1$$

The integral becomes:

$$\iint_S 1 \, dx \, dy = \text{Area of region } S$$

Since S is a circle of radius $R = 1$, its area is $\pi R^2 = \pi$. **RHS** = π

Since **LHS** = **RHS** = π , Green's Theorem is verified.

14.3.3 Problem 3: Verifying Green's Theorem between Curves

Question: Verify Green's Theorem for $\oint_C (xy + y^2) dx + x^2 dy$ where C is the closed curve bounded by $y = x^2$ and $y = x$.

Solution: The intersection points of $y = x^2$ and $y = x$ are $(0, 0)$ and $(1, 1)$.

Part 1: Line Integral (LHS) The boundary C has two paths traversed in an anticlockwise loop: 1. C_1

(Lower Curve): Along $y = x^2$ from $x = 0$ to $x = 1$. ($dy = 2x dx$)

$$I_1 = \int_0^1 [(x(x^2) + (x^2)^2) + x^2(2x)] dx = \int_0^1 (3x^3 + x^4) dx = \left[\frac{3}{4}x^4 + \frac{1}{5}x^5 \right]_0^1 = \frac{19}{20}$$

2. C_2 **(Upper Curve):** Along $y = x$ from $x = 1$ to $x = 0$. ($dy = dx$)

$$I_2 = \int_1^0 [(x^2 + x^2) + x^2] dx = \int_1^0 3x^2 dx = [x^3]_1^0 = -1$$

Summing the paths: $\text{LHS} = \frac{19}{20} - 1 = -\frac{1}{20}$.

Part 2: Double Integral (RHS)

$$P = xy + y^2 \implies \frac{\partial P}{\partial y} = x + 2y.$$

$$Q = x^2 \implies \frac{\partial Q}{\partial x} = 2x.$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - (x + 2y) = x - 2y$$

Set up the bounds: For a given $x \in [0, 1]$, y ranges from the lower curve ($y = x^2$) to the upper curve ($y = x$).

$$\begin{aligned} \text{RHS} &= \int_0^1 \int_{x^2}^x (x - 2y) dy dx = \int_0^1 [xy - y^2]_{x^2}^x dx \\ &= \int_0^1 [(x^2 - x^2) - (x^3 - x^4)] dx = \int_0^1 (x^4 - x^3) dx \\ &= \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20} \end{aligned}$$

Since $\text{LHS} = \text{RHS} = -\frac{1}{20}$, Green's Theorem is verified.

14.4 Conclusion

This marks the conclusion of our Vector Calculus series! From basic scalar and vector point functions, to the differential operators Gradient, Curl, and Divergence, and finally the monumental theorems of Stokes, Gauss, and Green that unify vector integration. We are now fully equipped to transition into Complex Analysis.